Control Is Motion

Geometry, Feasible Dynamics, and Robust Outcomes

Preface

This is Volume II of *The Geometry of Motion*. Where Volume 0 laid the kinematic and algebraic groundwork (rigid-body transforms, screw theory, spatial algebra, and the Lagrangian formulation) and Volume I developed the tangent-space machinery for linearization, contraction, and optimal control, this volume examines motion undertaken for a desired outcome. Athletic motions, legged locomotion, aerial maneuvers, and manipulation through contact require more than a static pose. Their objectives may combine time-indexed tracking, path following, terminal delivery, and orbital behavior. The appropriate model must account for configuration geometry, feasible inputs, contact, and relevant uncertainty. These questions extend established control analysis; they do not make targets or Lyapunov theory obsolete.

The volume is organized around three concentric questions. First, what is the *right* mathematical specification for the task? A path, timing law, terminal output set, and verified tube serve different purposes. Second, how do we *synthesize* compatible trajectories and policies? We develop trajectory optimization, funnel synthesis, and phase-variable control as complementary tools. Third, how do the resulting techniques connect to biological motor control and learning? We close with stochastic motor control and a proposed golf case study. A research proposal must still supply its full model, reproducible calculations, and independent validation before its predictions can establish how golfers move or control the swing.

Prerequisites. Readers are assumed to be comfortable with the material of Volume 0 (rigid-body kinematics, Lagrangian mechanics) and Volume I (linearization, LQR, contraction metrics). Some familiarity with differential geometry at the level of tangent spaces and vector fields is assumed; the nomenclature table and Volume I appendix provide the minimum required background.

Relationship to Volume III. Volume III will extend these trajectory-centric tools to networked and multi-agent systems. The connection between motion, constraints, and desired outcomes carries forward.

Contents

Nomenclature

General Conventions

- Scalars are lowercase or Greek letters (e.g., m, t, θ).
- Vectors are bold lowercase (e.g., $\mathbf{x}, \mathbf{u}, \mathbf{q}$).
- Matrices are bold uppercase (e.g., $\mathbf{A}, \mathbf{B}, \mathbf{M}, \mathbf{J}$).
- Expected values and sets are denoted by blackboard bold or script (e.g., $\mathbb{E}, \mathcal{S}$).

State and Kinematics

$\mathbf{q} \in \mathbb{R}^n$ Joint configuration vector (generalized coordinates).

$\dot{\mathbf{q}} \in \mathbb{R}^n$ Joint velocity vector (generalized velocities).

$\ddot{\mathbf{q}} \in \mathbb{R}^n$ Joint acceleration vector.

$\mathbf{x} \in \mathbb{R}^{nx}$ System state vector; commonly $\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$ for mechanical systems.

$\mathbf{u} \in \mathbb{R}^m$ Control input vector (e.g., joint torques, muscle activations).

$\boldsymbol{\tau} \in \mathbb{R}^n$ Generalized forces/torques acting on the joints.

$t \in \mathbb{R}$ Time.

Inertia and Dynamics

$\mathbf{M}(\mathbf{q})$ Mass (inertia) matrix, symmetric positive definite.

$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ Coriolis and centrifugal force matrix.

$\mathbf{g}(\mathbf{q})$ Gravity vector.

$\mathbf{J}(\mathbf{q})$ Jacobian matrix relating joint velocities to task-space velocities ($\dot{\mathbf{p}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$).

Control and Optimization

$\mathbf{A}_k = \frac{\partial f}{\partial \mathbf{x}}$ Local Jacobian matrix of the state dynamics.

$\mathbf{B}_k = \frac{\partial f}{\partial \mathbf{u}}$ Local Jacobian matrix of the control input.

$V(\mathbf{x})$ **or** $\mathcal{V}$ Value function or Lyapunov function.

$\mathbf{Q}$ State penalty matrix in LQR/DDP.

$\mathbf{R}$ Control penalty matrix in LQR/DDP.

$\mathbf{K}$ Feedback gain matrix ($\delta\mathbf{u} = -\mathbf{K}\delta\mathbf{x}$).

$\boldsymbol{\gamma}$ A trajectory or flow, mapping time and initial conditions to states.

Affine Control and Counterfactuals

$f(\mathbf{x})$ Drift vector field: complete autonomous evolution of the declared effective plant when $\mathbf{u} = \mathbf{0}$.

$G(\mathbf{x})$ Input (control) matrix field mapping control inputs to state derivatives.

$\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$ Control-affine system equations of motion.

$\mathbf{x}^{\mathrm{ZTCF}}(t)$ Forward Zero-Torque Counterfactual trajectory under $\mathbf{u}(t) \equiv \mathbf{0}$.

$\mathbf{a}_{\mathrm{ZVCF}}(\mathbf{q}, \mathbf{z})$ Instantaneous Zero-Velocity Counterfactual acceleration with velocity and declared control set to zero.

$\mathrm{DCR}_{W,\mathcal{U}}(\mathbf{x})$ Drift-Control Ratio in a declared acceleration or task-projected space: $\frac{\|W\mathbf{a}_{\mathrm{drift}}\|_2}{\sup_{\mathbf{u}\in\mathcal{U}} \|WB_a\mathbf{u}\|_2+\varepsilon}$.

Biomechanics and Motor Control

$a_i \in [0, 1]$ Muscle activation level.

ℓ_i^M Muscle fiber length.

v_i^M Muscle fiber contraction velocity.

ℓ_i^{MT} Musculotendon unit length.

$\mathbf{F_{muscle}}$ Force generated by muscles.

$\mathbf{W}$ Muscle synergy matrix (weights).

$\mathbf{c}(t)$ Muscle synergy activation vector over time.

Geometry and Manifolds

$SE(3)$ Special Euclidean group of rigid body motions.

$SO(3)$ Special Orthogonal group of 3D rotations.

$\mathfrak{se}(3)$ Lie algebra of $SE(3)$, space of spatial twists (velocities).

$\mathcal{M}$ A smooth manifold.

$T_{\mathbf{x}}\mathcal{M}$ Tangent space at point $\mathbf{x}$.

Chapter 1

Motion, Targets, and Timing

A golf swing is a motion undertaken for an outcome. The club must arrive at contact with a useful position, velocity, orientation, and impact location; the subsequent collision and flight determine what the ball does. The body must produce that delivery through a feasible movement, tolerate relevant variation, and continue into follow-through. Studying only a static pose misses much of this problem. Discarding the outcome would miss its purpose.

"Control is motion" is the organizing perspective of this book. It directs attention to how a trajectory is generated, shaped, and made reliable. It is not a claim that trajectory tracking, optimal control, or orbital stability replaces classical control theory. Those established tools already address moving references, periodic behavior, disturbances, and constrained objectives. The task is to choose and combine them correctly for the question at hand.

The important connections are physical. Earlier applied work changes the state available later. Configuration and velocity change inertial coupling, force transmission, and external loads. Grip and ground constraints restrict compatible motion and affect reactions. Timing changes velocity and effort even along an unchanged configuration path. Feedback and constitutive response alter how disturbances develop. Impact samples the resulting state at an event, which need not occur at a fixed clock time. A rigorous account keeps these connections while distinguishing a model identity from a claim about a golfer.

1.1 Specify What the Motion Should Produce

Let the state be x, the effective input be u, and the declared dynamics be

$$\dot{x} = f(x, u, t), \qquad u \in \mathcal{U}(x, t). \tag{1.1}$$

The state must contain enough information for the model to predict its own continuation. For a rigid mechanical model it includes configuration and velocity. A muscle-driven or flexible model may also need activation, tissue, shaft, and other internal states. A list of joint angles alone is not generally a complete dynamic state. Contact modes and their transition rules are part of the model as well.

Several objectives are useful and can coexist. Regulation keeps a quantity near a desired constant value. Trajectory tracking compares the state or output with a time-indexed reference. Path following permits progress along a geometric path with a separately specified timing objective. Terminal or event objectives constrain the state delivered at a particular time or crossing. Periodic tasks may call for orbital stabilization. The names describe different specifications; they are not mutually exclusive schools of control.

For a simplified impact investigation, let the first intended approach crossing be described by

$$h(x(t_e)) = 0, \qquad \nabla h(x(t_e))^T f(x(t_e), u(t_e), t_e) \neq 0, \tag{1.2}$$

with an appropriate approach direction and contact mode. The nonzero derivative is a transversality condition: it excludes a grazing crossing where event time can be especially sensitive. Define an output $y_e = H(x(t_e^-))$ from the pre-impact state and a target set $\mathcal{Y}$ of acceptable deliveries. Its components might describe clubhead velocity, face orientation, path direction, and impact location. A ball-outcome model can map those quantities, together with ball and contact properties, into launch and flight predictions.

A terminal requirement $y_e \in \mathcal{Y}$ is compatible with a moving club. It does not demand convergence to an equilibrium with nonzero velocity, nor require that the club accelerate through the collision. The impact model must account for the rapid change in momentum. Follow-through and any subsequent constraints require their own continuation. The event model, output definition, and admissible set determine what constitutes success.

This distinction is practical: two motions can have similar joint paths but different delivery speeds, while different joint paths can deliver similar club states. Neither observation by itself identifies how individual muscles produced the motion.

1.2 A Moving Reference Must Be Feasible

A nominal time-indexed trajectory $x_d(t)$ must have a compatible admissible input $u_d(t)$:

$$\dot{x}_d(t) = f(x_d(t), u_d(t), t), \qquad u_d(t) \in \mathcal{U}(x_d(t), t). \tag{1.3}$$

Joint limits, closure conditions, contact forces, and internal-state dynamics must also hold. Drawing a smooth curve in state space does not establish these properties. If a tracking error is expected to remain zero, the closed-loop vector field must reproduce the nominal tangent there.

For a control-affine model $f = f_0 + Gu$, define $r = \dot{x}_d - f_0(x_d, t)$. At a given nominal state, an unconstrained input exists only if $r \in \operatorname{im} G$. When that condition holds,

$$u_d = G^\dagger r + (I - G^\dagger G)z \tag{1.4}$$

expresses the family of solutions in chosen coordinates. The pseudoinverse selects a minimum Euclidean-norm representative; that criterion depends on input scaling and is not a muscle-recruitment law. Input bounds can eliminate all members of the family. A nonlinear input map may require a different solution procedure and still need not have an inverse.

For example, $G = (1, 0)^T$ cannot produce $r = (0, 1)^T$. Its least-squares solution leaves a nonzero residual. Conversely, $G = (1, 1)$ with $r = 1$ admits both $(1, 0)^T$ and $(1/2, 1/2)^T$, among infinitely many choices. Feasibility, uniqueness, and the selection of a preferred input are separate questions.

One useful architecture is $u = u_d + \Delta u$, with corrections designed for the actual tracking-error dynamics. It does not imply a universal feedforward share of measured joint torque. A nominal passive motion can have $u_d = 0$, whereas a strongly forced motion may require substantial nominal input. The split also depends on the chosen reference and policy representation. Establishing a neural feedforward or feedback contribution requires an experimental and physiological model beyond this algebraic split.

1.3 Tracking and Regulation

In a local Euclidean chart, set $e = x - x_d(t)$. Then

$$\dot{e} = f(e + x_d(t), u, t) - \dot{x}_d(t). \tag{1.5}$$

Tracking becomes regulation of the origin in a generally time-varying error system. This change of variables explains why Lyapunov methods and linear-quadratic designs remain useful for moving references. The MIT treatment of finite-horizon LQR and trajectory stabilization provides an established development of these tools. Applying an equilibrium controller without accounting for the changing reference can fail; that is a modeling or design error, not a limitation of all regulation-based analysis.

For a scalar example, take $\dot{x} = x + u$ and $x_d(t) = \sin t$. The nominal input is $u_d = \cos t - \sin t$. Choosing

$$u = u_d - (k+1)e, \qquad k > 0, \tag{1.6}$$

gives $\dot{e} = -ke$ exactly. The state keeps moving while its tracking error converges. If actuator bounds are imposed, this result applies only where the specified input can actually be delivered. Delays, unmodeled dynamics, and measurement error also require their own treatment.

1.4 Configuration Paths, Timing, and State

Write a configuration path as $q_d(s)$, where s is a dimensionless progress coordinate. For a timing law $s(t)$, define $\sigma = \dot{s}$. The chain rule gives

$$\begin{aligned} q(t) &= q_d(s(t)), \\ \dot{q}(t) &= q_d'(s)\sigma, \\ \ddot{q}(t) &= q_d''(s)\sigma^2 + q_d'(s)\dot{\sigma}. \end{aligned} \tag{1.7}$$

Thus the mechanical state is $(q_d(s), q_d'(s)\sigma)$, not a configuration-only curve. Changing σ changes that state and usually the required loads. A prescribed full-state curve $(q_d(s), v_d(s))$ has an additional compatibility condition $q_d'(s)\sigma = v_d(s)$. It cannot generally be traversed at any chosen speed while remaining the same physical state trajectory.

For a mechanical model $M(q)\ddot{q} + h(q, \dot{q}) = B(q)u$, substitution yields

$$B(q_d)u = M(q_d)\big(q_d''\sigma^2 + q_d'\dot{\sigma}\big) + h(q_d, q_d'\sigma). \tag{1.8}$$

The right side must lie in the admissible input image. Closure reactions, external forces, and constitutive loads must be included consistently in a more complete model. Geometric path design and timing design are useful subproblems, but the dynamics couples them. Underactuation makes that compatibility especially important; it does not make arbitrary retiming free.

As a reproducible example, take $m\ddot{q} = u$ with mass $m = 1$ kg and applied force u. Let $q_d(s) = \ell s^2$ with $\ell = 1$ m. Use $s = t/T$ on $0 \le t \le T$. The same configuration path from 0 to 1 metre gives

$$q = \ell(t/T)^2, \qquad \dot{q} = 2\ell t/T^2, \qquad u = 2m\ell/T^2. \tag{1.9}$$

At the endpoint, $T = 1$ s gives speed 2 m/s, force 2 N, and kinetic energy 2 J. With $T = 2$ s, the values are 1 m/s, 0.5 N, and 0.5 J. Work over the one-metre displacement equals the kinetic-energy increase in each case. The phase-plane curves are $\dot{q} = 2\sqrt{\ell q}/T$, so they are different state-space curves. This is an illustrative timing calculation, not a measured golf swing.

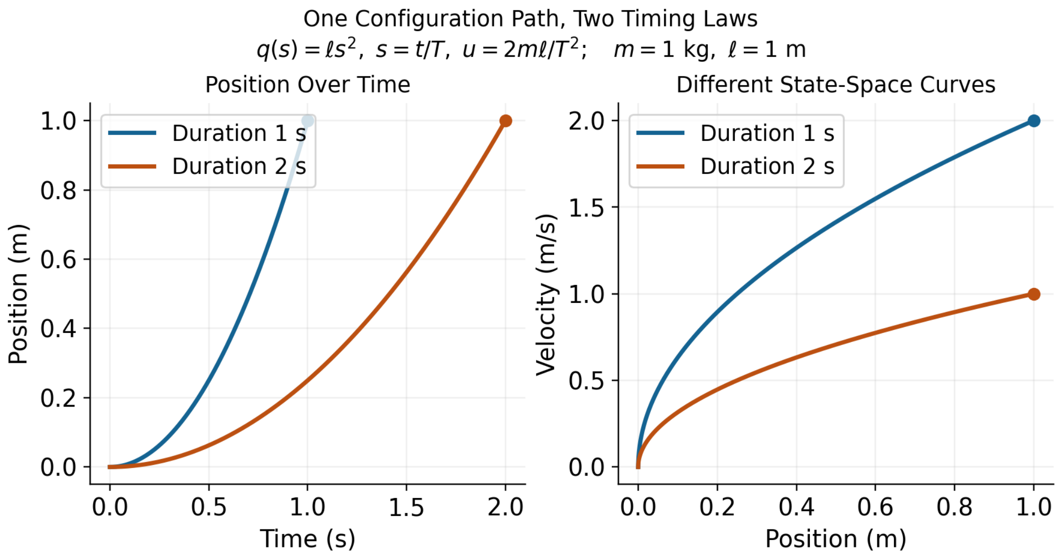


Figure 1.1: One Configuration Path, Two Timing Laws, and Different State Trajectories. Curves End at the Same Position With Different Velocities.

The example uses a prescribed clock-based phase. Estimating phase from the configuration itself needs care near the stationary start, where $q_d'(0) = 0$. In a swing, a coordinate that reverses at the top of the backswing cannot serve as a globally increasing phase through that reversal without a different construction. A useful phase must be defined on the intended interval, be observable from the chosen state or measurements, and have the required regularity and progress properties.

1.5 Choose a Meaningful Metric and Phase

A raw norm of position and velocity coordinates mixes different units. Even among positions, changing metres to centimetres changes an unweighted norm. One local choice is

$$\|e\|_W^2 = e^T W e, \qquad W = W^T \succ 0, \tag{1.10}$$

with weights and scales declared according to the desired interpretation. Under an invertible linear coordinate change $\widetilde{e} = Se$, the same quantity uses $\widetilde{W} = S^{-T} W S^{-1}$. Keeping W unchanged while changing units generally changes the specification. A physically motivated metric or scaled output error can be useful, but neither is automatically a clinical risk measure or an empirical measure of human effort.

On a configuration manifold, subtraction is a local coordinate operation; rotations, for example, need compatible charts or a chosen local logarithmic error. The mass matrix defines a kinetic-energy metric on configuration velocities, not by itself a unique distance on the entire augmented state including angles, velocities, activation, and other quantities.

For an embedded regular reference curve $\gamma(s)$ in a scaled state chart, one possible local phase is

$$s_*(x) \in \arg\min_s \frac{1}{2}(x-\gamma(s))^T W(x-\gamma(s)). \tag{1.11}$$

At an interior minimizer with constant W,

$$\gamma'(s_*)^T W(x-\gamma(s_*)) = 0. \tag{1.12}$$

A sufficiently small tubular neighborhood of a suitably regular embedded curve can make the projection unique and smooth. That conclusion is local. At the center of a circle every point on the circle is equally close; self-intersections, nearby branches, zero tangents, or a finite arc's endpoints can invalidate the assumed phase chart. A constrained endpoint minimizer need not satisfy the interior orthogonality equation.

Transverse coordinates need not use nearest-point orthogonality, but must form valid local coordinates together with phase. If the state dimension is d, there are locally $d-1$ transverse coordinates around a regular one-dimensional curve. This coordinate count is different from the number of actuators and from the dimension of a task-equivalent set.

1.6 Stability, Attraction, and Finite Motion

Lyapunov stability means sufficiently small initial errors remain small over the stated time domain. It does not by itself mean convergence. Asymptotic stability adds attraction as time tends to infinity. Exponential stability adds a quantitative decay bound with specified constants. These distinctions apply to equilibria and have corresponding trajectory and orbital formulations; see the MIT Lyapunov analysis notes.

For a periodic orbit of an autonomous closed-loop system, orbital stability allows phase displacement while controlling distance to the orbit. A nearby solution can converge to the orbit without converging to the same clock phase as a nominal solution. This is useful for periodic motions when phase locking is not the objective. It is not a general license to change the physical speed along a prescribed position-velocity curve.

A golf swing studied only up to an impact event is a finite-duration motion. Its appropriate claims may concern bounded deviation, successful event arrival, and delivered output tolerances. An infinite-time attraction statement needs a defined continuation, such as a periodic task or a subsequent stabilization phase. A finite reference arc is not an invariant orbit once the motion leaves its endpoint. Bounds on a finite interval should not be renamed asymptotic stability without that missing structure.

For example, $\dot{e} = e$ gives $e(t) = e(0)e^t$. On a fixed interval $[0, T]$, choosing $|e(0)| < \varepsilon e^{-T}$ keeps $|e(t)| < \varepsilon$. This finite-time bound holds even though errors amplify and the origin is unstable on the infinite time axis. A useful swing robustness result should therefore report the permitted initial/measurement uncertainty, the amplification or decay, the time or event interval, and the terminal output bound.

Orbital analysis still uses Lyapunov functions. For an explicit example, consider $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + u$. The unit circle is a nominal orbit with $u_d = 0$. Put $r^2 = x_1^2 + x_2^2$ and choose

$$u = kx_2(1-r^2), \qquad V = \tfrac{1}{4}(r^2-1)^2, \qquad \dot{V} = -kx_2^2(r^2-1)^2 \le 0, \quad k > 0. \tag{1.13}$$

On a sufficiently small compact annulus around the unit circle, the largest invariant subset of $\dot{V} = 0$ is the circle itself. Points with $x_2 = 0$ away from the circle do not remain there, since $\dot{x}_2 = -x_1 \neq 0$; the origin is excluded by the annulus. Lyapunov and invariance arguments then establish local orbital attraction and stability. They do not establish phase locking or global attraction from the origin. The MIT limit-cycle treatment and Manchester's transverse-dynamics analysis provide the broader framework.

1.7 Verifying a Motion Tube

A time-varying tube can be defined by

$$\mathcal{T}(t) = \{x : V(x,t) \leq \rho(t)\}. \tag{1.14}$$

For example, V may measure weighted tracking error. Selecting $\rho(t)$ to be small near impact expresses a desired tolerance. It does not show that the closed-loop motion can stay inside the tube. Even $\dot{e} = 0$ leaves a shrinking tube when initialized on its boundary: constant error cannot fit inside an arbitrarily decreasing allowed radius.

Suppose the closed-loop dynamics is well posed with admissible inputs, V and ρ are continuously differentiable, and the tube has a regular boundary on the domain of interest. For disturbances d in a declared set $\mathcal{D}$, a robust inward-boundary condition is

$$\partial_t V + \nabla_x V^T f(x, k(x,t), d, t) \leq \dot{\rho}(t) \quad \text{when } V(x,t) = \rho(t), \quad \forall d \in \mathcal{D}. \tag{1.15}$$

Together with the appropriate regularity and initial membership, this supports forward invariance over the specified interval. The derivative must include reference motion and changing metric weights. If V and ρ are indexed by phase, their derivatives involve the actual $\dot{s}$; it cannot silently be replaced by one. Nonsmooth profile corners or hybrid resets need their own one-sided or transition conditions.

A scalar example makes the effort and disturbance limit explicit. Let $\dot{e} = -ke + d$, $|d| \leq \bar{d}$, and require $|e| \leq r(t)$. On either boundary, a sufficient condition is

$$\dot{r} \geq -kr + \bar{d}. \tag{1.16}$$

For $k = 2$, $\bar{d} = 0.1$, and $r(0) = 1$ in declared scaled units, the tight comparison radius is $r(t) = 0.05 + 0.95e^{-2t}$. Constant worst-direction disturbances attain its two boundaries. The nonzero limiting radius follows from the disturbance bound and available correction rate. A drawing that shrinks to zero would not certify this disturbed system.

For an impact specification, also require that every relevant pre-impact boundary state allowed by the verified tube maps into $\mathcal{Y}$, and that the intended event is reached. A phase estimate bounded below by $\dot{s} \geq \alpha_{\min} > 0$ can supply a progress bound on an appropriate interval; distance to a path alone cannot. Actuator saturation, sensor delay, contact feasibility, and uncertainty in the model must be included in the claimed guarantee. A verified model tube is still a statement about that model and its declared uncertainty set, not automatic validation on golfers.

1.8 Input Rank and Reachability

For an n-coordinate mechanical model with $m < n$ effective inputs, rank limits instantaneous input acceleration directions. It does not imply that all feasible states lie on a surface of dimension $2m$. In a state model, drift and input directions evolve as the system moves; their effect over time can provide additional reachable directions. The underactuation chapter develops the associated feasibility and reachability questions.

A concrete counterexample uses two unit masses, one grounded spring, and one coupling spring, all with unit coefficients. In scaled coordinates,

$$\ddot{q} + Kq = bu, \qquad K = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{1.17}$$

There is one applied force. With $x = (q_1, q_2, \dot{q}_1, \dot{q}_2)^T$,

$$\dot{x} = Ax + gu, \qquad A = \begin{pmatrix} 0 & I \\ -K & 0 \end{pmatrix}, \qquad g = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}. \tag{1.18}$$

The controllability matrix $[g, Ag, A^2g, A^3g]$ has determinant -1 and rank four. With unconstrained signed inputs, this linear system is controllable in its full four-dimensional state space. Input limits restrict the size of reachable sets and time needed; they do not restore the asserted universal $2m$ dimension bound.

A single trajectory is itself a one-dimensional curve, but the union of reachable trajectories and the span of a curve's successive derivatives are different objects. For the same system with $u = 0$ and $x(0) = (1, 0, 0, 0)^T$, the matrix $[Ax, A^2x, A^3x, A^4x]$ at the initial state also has determinant -1. Thus even an unforced example has four independent time derivatives. At that regular point, smooth arc-length reparameterization preserves their rank. Actuator count therefore does not cap the number of independent Frenet-frame directions at $2m$ either.

The useful golf interpretation is more specific. Coupling can transmit and redistribute the effects of earlier work and present loads. Whether an effective wrist coordinate is unactuated, weakly actuated, or actively constrained is a modeling and measurement question. A rapid release does not prove zero wrist moment, and passive planar dynamics does not guarantee three-dimensional face squaring. Force, joint moment, mechanical power, and muscle activation must be kept distinct when testing such hypotheses.

1.9 Reliability and Performance

A path error, a delivery error, and a ball-dispersion measure are different costs. Choose the one that represents the study, then state how the others enter the model. Both regulation and trajectory problems can use cost functionals over an entire history; neither is restricted to a single instantaneous error value.

For a specified stochastic model and admissible policy π, one possible criterion is

$$\begin{aligned} J_\pi &= \phi(y_e) + \int_0^{t_e} \ell(x(t), u(t))\, dt, \\ \min_\pi \quad & \mathbb{E}[J_\pi] + \lambda \operatorname{Var}(J_\pi), \qquad \lambda \geq 0. \end{aligned} \tag{1.19}$$

The cost and its units, disturbance distribution, event definition, and constraint handling must be declared. The variance weight has corresponding units; it expresses a design preference rather than a universal biological constant. A low variance can accompany a consistently poor mean outcome, so repeatability alone is not sufficient. Expected cost, tail risk, target-set probability, and worst-case guarantees answer different questions. If the intended event is not reached by an allowed horizon, define a failure cost or constraint; silently discarding those trials changes the question.

A candidate policy may use feedback, a learned reference, or both. Comparing policies requires recomputing the motion, loads, event time, and output in each branch. Reusing a measured state or reaction history after the intervention changes the trajectory can invalidate the comparison. Experimental perturbations and held-out predictions can test a model; fitting one observed motion does not establish that the nervous system optimized the proposed criterion.

A practical golf investigation can proceed as follows:

1. Define the delivered output, its tolerance, and the ball/contact model used to connect that output to performance.
2. Declare the mechanical boundary, contact modes, state variables, input meaning, parameter uncertainty, and available measurements.
3. Construct compatible nominal states and inputs. Check closure, timing, actuator limits, work, and contact feasibility before optimizing a path.
4. Model disturbances and measurement/actuation delays. Compare finite-time or event-based output sensitivity and verify any proposed tube condition.
5. Test predictions on independent swings or perturbations, including uncertainty in event timing and the inferred loads.
6. Separate demonstrated mechanical identities, validated predictions, physiological interpretations, and proposed coaching or design hypotheses.

This is the role of the subsequent chapters: curve geometry describes motion; configuration manifolds describe compatible coordinates; transverse analysis and underactuation constrain candidate designs; optimization and robustness analysis evaluate them; learning and experiments test how well they transfer. The later golf case study is a modeling proposal unless supported by a complete specified model, reproducible results, and validation data. The framework's value comes from making the links testable.

1.10 Chapter Exercises

1. Specify an impact-event output set that distinguishes clubhead speed, face orientation, and impact location. Explain why a terminal target does not require stopping there.
2. Derive the error dynamics for the scalar tracking example, then impose an input bound and identify where the proposed law becomes infeasible.
3. Reproduce the one-mass timing example. Compare endpoint velocity, input, work, and the phase-plane curves for $T = 1$ and $T = 2$ seconds.
4. For a prescribed pair $(q_d(s), v_d(s))$, derive the compatibility condition on $\dot{s}$ and give an example where arbitrary retiming fails.
5. Rescale a position coordinate from metres to centimetres. Transport the error metric and verify that the weighted distance remains unchanged.

6. Show why nearest-point phase is not unique at a circle's center and why a finite arc's endpoint needs a separate projection condition.

7. Derive the scalar robust-radius condition and integrate its two worst-direction disturbance cases. Explain the nonzero limiting radius.

8. Verify both rank-four matrices in the spring example. Distinguish actuator rank, reachable-set dimension, and a trajectory's derivative span.

9. Verify the orbital Lyapunov derivative and the invariant-set argument. Explain why this does not prove phase locking or global convergence.

10. Formulate two different reliability objectives for a golf study. State what data would distinguish their predictions and what remains unidentifiable.

Chapter 2

Curves in State Space

2.1 What a Curve Describes

A golf swing has several useful geometric descriptions. The clubhead traces a curve in three-dimensional physical space. The joint configurations trace a curve in configuration space. Positions, velocities, and any additional internal states trace a curve in state space. These descriptions answer different questions. A bend in a plot of joint angle against angular velocity is not a bend of the clubhead in metres, and its curvature is not a force.

The purpose of this chapter is to make geometry useful without asking it to supply missing dynamics. We develop arc length, curvature, moving frames, local transverse coordinates, and timing constraints. We then connect these tools to physical acceleration, admissible control, and the impact task. Geometry specifies a motion's shape within a stated space and metric. Dynamics determines which timed motions and corrections are possible. Measurements determine whether a proposed model describes an actual swing.

Throughout the Euclidean calculations, the ambient space is $\mathbb{R}^d$ with a specified inner product. A prime denotes differentiation with respect to the displayed curve parameter; a dot denotes physical time. A regular curve has a nonzero first derivative on the interval under discussion. Configurations are denoted by q, full states by x, and a physical point such as the clubhead by p. We use s for arc length and λ for a parameter that need not measure distance.

2.1.1 Geometry Depends on the Space and Metric

Euclidean curvature is invariant under regular reparameterization and rigid changes of Cartesian frame. It is an extrinsic property of the embedded curve: it measures how the curve bends in its ambient space. Calling this "intrinsic" merely because it does not depend on timing confuses two uses of geometry. A straight segment and a circular arc of the same length have the same one-dimensional arc-length metric, although their ambient curvatures differ.

Changing units without transforming the metric changes the calculation. For a state $x = (q, \dot{q})$, a raw sum of squared positions and velocities has no universal physical meaning. One possible declared convention is $z = Sx$, where S contains inverse reference scales, followed by Euclidean geometry in the dimensionless coordinates z. Different scales generally give different curvature signatures. This is a modeling choice to report and test, not a coordinate-free marker of skill.

For a positive-definite metric W, the squared displacement is $dx^T W dx$. Under $z = Sx$, the same metric has matrix $\widetilde{W} = S^{-T} W S^{-1}$. A constant metric can be handled through a linear change to orthonormal coordinates. With a position-dependent metric, ordinary second derivatives are not sufficient: curvature uses the appropriate covariant derivative. The kinetic-energy matrix $M(q)$ defines a metric on configuration velocities; it is not automatically a metric on an augmented state containing velocities, activation, and contact variables.

2.2 Regular Parameters and Arc Length

Let $\gamma : I \to \mathbb{R}^d$ be a C^2 curve, and let $h(\lambda) = \|\gamma'(\lambda)\| > 0$. Its arc-length function and unit tangent are

$$s(\lambda) = \int_{\lambda_0}^{\lambda} h(\mu)\, d\mu, \qquad T(\lambda) = \frac{\gamma'(\lambda)}{h(\lambda)}. \tag{2.1}$$

Because $ds/d\lambda = h > 0$, the inverse function theorem gives a local regular arc-length parameter. In that parameter $\|\gamma'(s)\| = 1$ and $T = \gamma'(s)$. Replacing λ by a smooth increasing diffeomorphism changes the rate of traversal while preserving the oriented curve. A merely increasing function can have zero derivative and need not preserve regularity at every point.

For example, $(\cos t, \sin t)$ and $(\cos t^2, \sin t^2)$ trace the same circle on suitable intervals. The second time parameterization has zero velocity at $t = 0$. Arc-length formulas that divide by speed apply for $t > 0$; the circle's geometry can separately be extended through the starting point. A temporal stop is not by itself a geometric cusp or an infinite curvature.

A reversal is even more informative. In normalized coordinates, the physical path $p(t) = (t^2, 0)$ has zero curvature on either side of $t = 0$, while velocity vanishes and reverses there. The unit tangent defined by time is undefined at the reversal. The full mechanical state $x(t) = (t^2, 2t)$ has derivative $(2t, 2)$ and is regular at the same instant. Thus the physical point can stop while the state continues moving. There is no universal conclusion that the top of a backswing is the point of maximum curvature in any of these spaces.

If a regular configuration path is parameterized by λ and traversed with $\dot{\lambda} > 0$, its chosen-metric speed is $\dot{s} = \|q'(\lambda)\|\dot{\lambda}$. In particular, $\dot{\lambda}$ is not Cartesian speed unless the path is parameterized by physical arc length. This distinction matters when converting a speed limit into a timing law.

2.3 Curvature and Physical Acceleration

For a unit-speed C^2 Euclidean curve,

$$k(s) = \frac{dT}{ds}, \qquad \kappa(s) = \|k(s)\|. \tag{2.2}$$

Differentiating $T^T T = 1$ gives $T^T k = 0$. Where $\kappa > 0$, the principal normal is $N = k/\kappa$. At a point with zero curvature, k remains defined but this particular normal does not. Curvature vanishing on an entire connected regular interval makes T constant and the curve a straight segment. Vanishing at one point does not: (t, t^3) has zero curvature at $t = 0$ and bends arbitrarily nearby.

For any regular parameter, differentiating the normalized tangent gives

$$\kappa = \frac{\|(I - TT^T)\gamma''\|}{\|\gamma'\|^2} = \frac{\sqrt{\|\gamma'\|^2\|\gamma''\|^2 - (\gamma'^T\gamma'')^2}}{\|\gamma'\|^3}. \tag{2.3}$$

In three dimensions the numerator of the final expression is $\|\gamma' \times \gamma''\|$. In two dimensions it is the absolute determinant of the two derivative vectors. There is no need to invent an ordinary cross product in arbitrary dimension. The projected-acceleration form also avoids subtracting two large nearly equal squared quantities, although numerical error can still dominate near zero speed.

2.3.1 A Checkable Ellipse

For $\gamma(\lambda) = (a\cos\lambda, b\sin\lambda)$ with $a > b > 0$,

$$\kappa(\lambda) = \frac{ab}{(a^2\sin^2\lambda + b^2\cos^2\lambda)^{3/2}}. \tag{2.4}$$

The major-axis endpoints $(\pm a, 0)$, at $\lambda = 0, \pi$, have the largest curvature a/b^2. The minor-axis endpoints $(0, \pm b)$ have the smallest curvature b/a^2. With $a = 2$ m and $b = 1$ m these are 2 and 0.25 inverse metres. Uniformly changing the rate of traversal preserves these values. Stretching a circle into this ellipse does not preserve curvature.

2.3.2 Where the Speed-Squared Law Applies

Let $p(s)$ now be a physical position in an inertial Cartesian frame, with s measured in metres, and let $v = \dot{s} > 0$. The chain rule gives

$$\dot{p} = vT, \qquad \ddot{p} = \dot{v}T + v^2k = \dot{v}T + v^2\kappa N \quad (\kappa > 0). \tag{2.5}$$

Here $v^2\kappa$ is a physical normal acceleration in m/s^2. The identity is kinematic: it describes the acceleration of the prescribed motion before assigning which forces produce it.

For a point mass, Newton's law says

$$m\ddot{p} = F_{\text{act}} + F_{\text{pass}} + F_{\text{external}} + F_{\text{reaction}}. \tag{2.6}$$

Only in a model where the actuator supplies the entire resultant force does $F_{\text{act}} = m(\dot{v}T + v^2k)$. Gravity, springs, contact, and constraint reactions can contribute. A 0.2 kg mass moving at 20 m/s on a horizontal circle of radius 1 m requires 80 N inward. In an ideal frictionless circular guide with no tangential resistance, the guide supplies that force and does zero work because its force is perpendicular to velocity. No continuous actuator work is needed to maintain the speed in this model. That calculation does not show that a golfer supplies zero joint moments. It shows why resultant acceleration and active effort are different quantities.

A clubhead is a point on a moving, rotating, possibly flexible body rather than an isolated point mass with all force applied at that point. For a rigid model with $p = p(q)$,

$$\dot{p} = J_p(q)\dot{q}, \qquad \ddot{p} = J_p(q)\ddot{q} + \dot{J}_p(q, \dot{q})\dot{q}. \tag{2.7}$$

The Jacobian-rate term supplies normal acceleration even at constant joint speed. Segment force and moment balances, distributed mass, hand wrenches, and any modeled shaft deflection are needed to interpret measured clubhead acceleration. Multiplying it by total club mass does not generally recover the hand force.

For a first-order state equation $\dot{x} = f(x, u, t)$, the analogous curve identity concerns $\ddot{x}$, which may include mechanical jerk and internal state derivatives. When u is differentiable,

$$\ddot{x} = f_x f + f_u\dot{u} + f_t. \tag{2.8}$$

This is not a generic decomposition of u into tangential and normal forces. The reference condition remains $f(\gamma(s(t)), u_*(t), t) = \gamma'(s(t))\dot{s}(t)$, together with input admissibility. In an autonomous model the final argument is absent.

2.4 Moving Frames and Their Limits

2.4.1 An Oriented Three-Dimensional Frame

For a C^3 regular curve in oriented Euclidean three-space with $\kappa > 0$, define $T = dp/ds$, $N = T'/\kappa$, and $B = T \times N$. Then

$$\frac{dT}{ds} = \kappa N, \qquad \frac{dN}{ds} = -\kappa T + \tau B, \qquad \frac{dB}{ds} = -\tau N. \tag{2.9}$$

The signed torsion is

$$\tau = \frac{\det[p', p'', p''']}{\|p' \times p''\|^2}, \tag{2.10}$$

where the primes in this formula may use any regular parameter. Torsion requires nonzero curvature. A planar regular curve has zero torsion where the formula is defined; at an inflection the Frenet normal and torsion may be undefined even when the original curve is smooth.

For the helix $p(\lambda) = (a\cos\lambda, a\sin\lambda, b\lambda)$ with $a > 0$,

$$\kappa = \frac{a}{a^2 + b^2}, \qquad \tau = \frac{b}{a^2 + b^2}. \tag{2.11}$$

Changing the sign of b reflects the helix and changes the sign of torsion. Proper rotations and translations preserve both. Blindly applying Gram–Schmidt to all three derivatives chooses the final vector from the third derivative's residual; it can produce $-B$ when torsion is negative. That convention cannot simultaneously be identified with the oriented binormal $T \times N$ without tracking the sign.

In this physical three-dimensional setting, torsion measures rotation of the osculating plane per unit distance. Torsion of a normalized state-space curve has a different ambient space and interpretation. It is not a direct measure of anatomical rotation or of how a golfer should change the swing plane.

2.4.2 Higher Dimensions and Regular Normal Frames

With sufficient smoothness and independent successive derivatives, Gram–Schmidt constructs an orthonormal basis for the successive osculating spaces of a curve in $\mathbb{R}^d$. A full oriented Frenet frame can be chosen by orthogonalizing the first $d-1$ independent derivatives and completing the last vector with a fixed ambient orientation. Its equations have the form

$$e_i' = -\kappa_{i-1} e_{i-1} + \kappa_i e_{i+1}, \qquad \kappa_0 = \kappa_d = 0. \tag{2.12}$$

The first $d-2$ curvatures are positive under the required independence conditions; the final oriented curvature can have either sign or vanish. Alternatively, a frame made from all d independent derivatives adopts a different sign convention. State the convention before comparing signatures. If only a partial frame is constructed, its last derivative need not remain inside that partial span. A closed lower-dimensional Frenet system needs an appropriate containing affine subspace and its own regularity assumptions.

There is a simpler option for local control: choose any smooth orthonormal normal frame $E(s)$, with $d-1$ columns, satisfying $T^T E = 0$ and $E^T E = I$. Define

$$c = E^T T', \qquad \Omega = E^T E', \qquad T' = Ec, \qquad E' = -Tc^T + E\Omega. \tag{2.13}$$

Differentiating $E^T E = I$ shows that Ω is skew-symmetric. A parallel normal frame, often called a Bishop frame, chooses $\Omega = 0$ locally along an interval. It continues through zero-curvature points of a regular curve where the principal Frenet normal fails. Such a frame still requires an initial normal orientation; a periodic curve can have a nontrivial change of normal orientation after one circuit.

The number of normal coordinates is $d-1$, not the actuator count. The one-input spring example in the previous chapter has four independent unforced trajectory derivatives at its specified initial state. A single trajectory is one-dimensional, its osculating span can be larger, and the union of reachable trajectories is another object again.

2.5 Shape Reconstruction and Its Limits

For continuously differentiable $\kappa(s) > 0$ and continuous signed $\tau(s)$ on an interval, the three-dimensional Frenet equations, together with $p' = T$, reconstruct a unit-speed curve once its initial point and oriented frame are specified. Existence and uniqueness follow from the frame's linear differential equation and integration of T. Without the initial point and frame, the curve is determined up to a proper rigid motion. Analogous statements in higher dimensions require a complete consistent curvature list and frame convention; an arbitrary truncated signature is insufficient.

Congruence is not equality relative to the golf ball. Translating or rotating a clubhead path can make it miss the ball while preserving its curvature and torsion. A rigid rotation of normalized position-velocity coordinates need not even correspond to a physically allowed state transformation. Gravity, contact, anatomy, and the target frame further restrict useful symmetries. Neither shape reconstruction nor a similar curvature signature establishes equal timing, forces, face orientation, or performance.

Where $\kappa > 0$, the osculating circle has center $p + N/\kappa$ and radius $1/\kappa$. It matches local position, tangent, and curvature; the tangent line and osculating plane supply lower-order geometric descriptions. An osculating sphere in three dimensions needs additional conditions. Writing $\rho = 1/\kappa$, when $\tau \neq 0$ its center is $p + \rho N + (\rho'/\tau)B$. This follows by requiring successive derivatives of squared distance to a fixed center to vanish through third order. At zero torsion the construction may fail or be nonunique, depending on ρ'. It is not an unconditional approximation for every high-dimensional state curve.

2.6 Feasible Timing and Available Forces

For a chosen configuration path $q = q(\lambda)$,

$$\dot{q} = q'\dot{\lambda}, \qquad \ddot{q} = q''\dot{\lambda}^2 + q'\ddot{\lambda}. \tag{2.14}$$

Substitution into mechanical dynamics gives the required generalized load:

$$M(q)(q'\ddot{\lambda} + q''\dot{\lambda}^2) + h(q, q'\dot{\lambda}) = B(q)u + Q_{\text{pass}} + Q_{\text{external}} + J_c^T \eta. \tag{2.15}$$

Here h contains the declared velocity and potential terms, and passive terms must not be counted twice. The path must satisfy configuration constraints; its derivatives must satisfy their velocity and acceleration conditions. Contact reactions η, friction, actuator bounds, activation dynamics, and any force-rate limits constrain feasible timing. For an underactuated system, not every desired generalized load lies in the input image even if every motor is individually below its limit.

For a fully actuated model, each torque bound can become an inequality in λ, $\dot{\lambda}$, and $\ddot{\lambda}$. This is the basis of path-constrained time scaling. It is more informative than assigning a single curvature cost because it retains gravity, inertia, coupling, and the available input directions. The configuration path and timing can be optimized in stages, but their feasibility constraints remain coupled. Retiming also changes the corresponding full state curve, as shown in the previous chapter.

A simple point-mass model gives a useful limited speed bound. Suppose the actuator supplies the entire force, with $\|F\| \leq F_{\max}$, and there are no other forces. On a physical arc-length path,

$$\dot{v}^2 + (\kappa v^2)^2 \leq (F_{\max}/m)^2. \tag{2.16}$$

Thus $v \leq \sqrt{F_{\max}/(m\kappa)}$ is necessary when $\kappa > 0$, but leaves no force for tangential acceleration at equality. For $m = 2$ kg, $F_{\max} = 10$ N, $\kappa = 0.5\ \mathrm{m}^{-1}$, and $v = 3$ m/s, the normal acceleration is 4.5 m/s^2 and the remaining tangential bound is $\sqrt{25 - 4.5^2} = \sqrt{4.75}$ m/s^2. If a guide supplies normal reaction, the actuator's feasible set is different.

At zero curvature this particular normal-acceleration bound supplies no speed ceiling. Tangential acceleration, initial and terminal conditions, velocity limits, power, and the next bend can still limit speed. A minimum-time exercise based only on normal acceleration can therefore lack an attained smooth solution or a physically meaningful optimum. Prescribe the full bounds and endpoint conditions before claiming an optimal law.

2.7 Local Phase and Transverse Dynamics

2.7.1 Coordinates in a Tubular Neighborhood

Let $\gamma(s)$ be a regular embedded unit-speed state curve in a chosen Euclidean metric, and choose the smooth normal frame $E(s)$ above. In a sufficiently small neighborhood of an interior segment, write

$$x = \gamma(s) + E(s)\xi. \tag{2.17}$$

Nearest-point phase satisfies $T^T(x - \gamma) = 0$ there. Local uniqueness requires more than this stationary equation: the curve must not intersect itself in the neighborhood, and the coordinate Jacobian must be invertible. The scalar factor $1 - c^T\xi$ in that Jacobian must be nonzero. A smaller neighborhood may be needed to exclude other closest points. For a circle, the center has many nearest points and the inward normal coordinates become singular there. Endpoints require one-sided treatment. A regular local normal frame does not turn all nearby states into a single global chart.

Let the smooth autonomous dynamics be $\dot{x} = f(x, u)$ and suppose a feasible nominal input satisfies $f(\gamma(s), u_*(s)) = v_*(s)T(s)$ with $v_* > 0$. Differentiate the coordinate map and project onto T and E:

$$\dot{s} = \frac{T^T f(x,u)}{1 - c^T\xi}, \qquad \dot{\xi} = E^T f(x,u) - \Omega\xi\dot{s}. \tag{2.18}$$

These equations retain both the phase denominator and the normal-frame transport. They are coordinate-rate equations; the word "force" should not be assigned to every transport term in a first-order state model.

With $u = u_*(s) + \delta u$, the transverse linearization at $\xi = 0$ is

$$\dot{\xi} = A_\perp(s)\xi + B_\perp(s)\delta u + O(\|(\xi, \delta u)\|^2), \tag{2.19}$$

where

$$A_\perp = E^T f_x(\gamma, u_*)E - v_*\Omega, \qquad B_\perp = E^T f_u(\gamma, u_*). \tag{2.20}$$

All partial derivatives here hold s fixed; nominal phase evolves by $\dot{s} = v_*(s)$. For explicitly time-dependent dynamics, include the time state or derive the corresponding time-dependent coordinate map. A feedback depending on an independently prescribed clock is also a different problem from the phase-indexed law assumed here.

There is no universal correction of the form speed squared times a curvature matrix subtracted from the original Jacobian. Curvature enters the phase map and chosen frame; transverse stability depends on the actual vector field, its derivatives, input directions, feedback, and regularity. Changing the normal basis changes the coordinate matrices consistently, without changing physical attraction of the orbit.

2.7.2 The Same Circle With Opposite Stability

In a neighborhood of $r = R > 0$, consider the planar systems

$$\dot{r} = a(r - R), \qquad \dot{\theta} = \omega > 0. \tag{2.21}$$

Every value of a has exactly the same nominal circle, curvature $1/R$, speed $R\omega$, and nominal time parameterization. Yet radial error evolves as $e(t) = e(0)e^{at}$: $a < 0$ attracts, $a > 0$ repels, and $a = 0$ is neutral. Curvature and nominal speed cannot distinguish these cases. The figure uses $R = 2$ and $\omega = 1.3$, with initial radii 1.8 and 2.2, over four model time units. Its radial growth rates are $a = -0.3$ and $a = 0.3$.

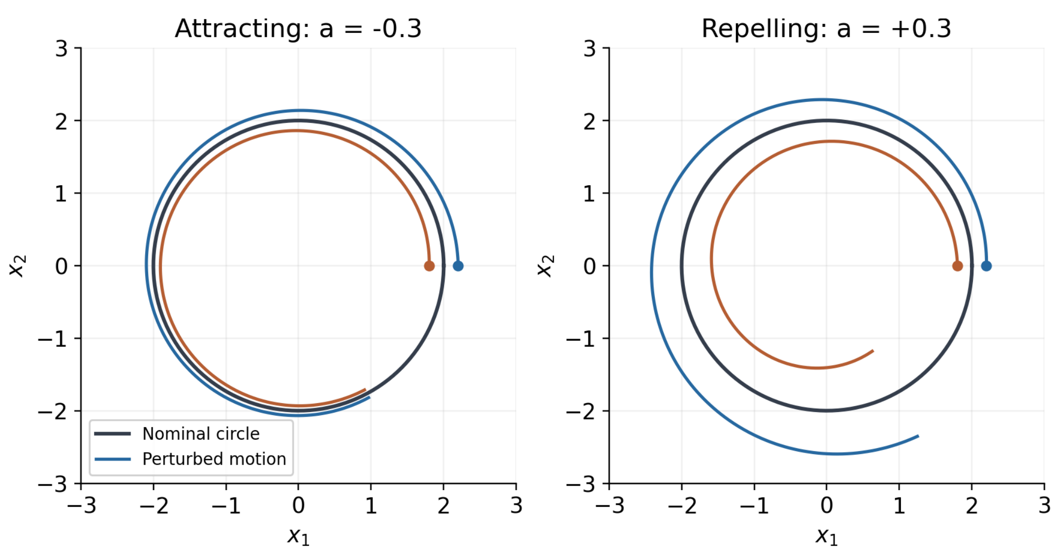


Figure 2.1: The Same Nominal Circle With Opposite Transverse Stability

Using the inward normal coordinate $\xi = R - r$ and arc length $s = R\theta$ gives $\dot{\xi} = a\xi$ and $\dot{s} = R\omega$. The exact phase formula agrees: $T^T f = \omega(R - \xi)$ and $1 - c^T\xi = 1 - \xi/R$. Their ratio is $R\omega$. This checks the denominator as well as the transverse linearization. A geometric tube around the circle becomes a verified invariant tube only when the actual transverse dynamics and disturbances satisfy its boundary condition.

2.8 Curvature and Reachability Are Different

For a control-affine system, instantaneous input directions, reachable sets, and the derivatives of one trajectory are distinct. Lie brackets describe noncommuting flows. With the convention $[g_i, g_j] = Dg_j\, g_i - Dg_i\, g_j$, the bracket can expose a displacement direction generated by suitable short sequences of controls. The Chow–Rashevskii result concerns bracket-generating driftless systems under appropriate smoothness and reversible control assumptions. The orbit theorem describes the geometry of orbits of vector fields and is not the same theorem. Allowing forward and backward field flows to define an orbit does not automatically describe forward-time reachability with drift, unilateral contact, or bounded controls.

A curved path need not use nonzero brackets at all. For the planar integrator $\dot{x} = u$, the two constant input fields commute. The control $u(t) = (-\sin t, \cos t)$ traces the unit circle from $(1, 0)$ with curvature one. Its acceleration comes from changing the input, while every bracket of the constant input fields is zero. Conversely, a system can have useful bracket directions without expressing them as a large curvature along one recorded trajectory. High curvature does not identify a commutator, a neural synergy, or an underactuated strategy.

2.9 Using Geometry in a Golf Investigation

A useful investigation keeps the chain of interpretation explicit: measurement, geometric description, feasible mechanics, uncertainty, and task outcome. Consider the following protocol.

1. Choose the object and frame. A clubhead point, shaft orientation, joint configuration, and complete state are different data products. Use a target-fixed physical frame for ball delivery. State how angles, rotations, velocities, and internal states are represented.

2. Report units and metric scales. For a state signature, repeat the calculation under plausible scaling choices. For three-dimensional clubhead curvature, use physical positions in consistent length units.

3. Fit derivatives with an explicit measurement model. Differentiation amplifies noise; torsion requires a third derivative and is especially fragile near small curvature or speed. Report sampling, smoothing, uncertainty, and sensitivity to filtering. Do not divide by near-zero denominators and then interpret numerical spikes as swing phases.

4. Align trials using a declared event or phase convention. Compare both geometric shape and timing; report where a reference curve's nearest projection becomes ambiguous. Treat impact and contact changes using one-sided smooth segments or a hybrid model.

5. Compute physical acceleration and mechanical load with the appropriate Jacobians, inertias, external loads, and constraints. Distinguish net joint moments from muscle recruitment and resultant force from work. Validate these quantities against independent measurements where possible.

6. Evaluate pre-impact speed, face orientation, path direction, strike location, and relevant ball outcomes with uncertainty. Test whether a geometric feature predicts outcomes beyond simpler measured variables on held-out trials. A correlation does not identify the control policy.

7. Test interventions in a declared forward model and then seek empirical validation. Altering timing can change speed-dependent loads and contact feasibility; altering a path can change moment arms and the impact frame. Recompute the complete trajectory rather than attaching a new conclusion to an unchanged force history.

The hypothesis that golfers exploit passive coupling remains worth testing. The geometric tools can identify repeatable patterns and organize model comparisons. They cannot by themselves establish that a particular pause, release, curvature peak, or low-dimensional signature is optimal, automatic, or a consequence of a particular neural strategy. Those arguments require mechanics and evidence at the next links in the chain.

2.10 Further Reading

The geometric conventions can be checked against Petersen's Classical Differential Geometry, especially its curve, Frenet, and parallel-frame treatments. Path-constrained timing and torque bounds are developed in Lynch and Park's Modern Robotics. For transverse dynamics and orbital stabilization, see Manchester's Transverse Dynamics and Regions of Stability for Nonlinear Hybrid Limit Cycles. These are mathematical and robotics references; they do not validate the golf-specific hypotheses in this chapter.

2.11 Exercises

1. For (t, t^3), derive the curvature and show why its zero value at the origin does not imply a straight neighborhood. Contrast this with the stopped physical path $(t^2, 0)$ and its regular position-velocity state.
2. Derive the ellipse formula and identify its major- and minor-axis endpoint curvatures. Verify invariance under $t \mapsto 2t$ and explain why stretching only one coordinate changes the Euclidean answer.
3. Compute curvature and signed torsion for (t, t^2, t^3) at $t = 0$ and $t = 1$. Explain what extra mechanical assumptions are needed to compare actuator effort along the two portions of the curve.
4. For the helix $(2\cos t, 2\sin t, t)$, calculate T, N, B, κ, τ. Reflect its third coordinate and track the sign of the oriented torsion.
5. Derive the exact normal-coordinate equations by projecting the derivative of $x = \gamma(s) + E(s)\xi$. Identify the coordinate singularity for a circle and verify the radial stability counterexample.

6. A 2 kg point mass is driven solely by a force of norm at most 10 N. Find the available tangential acceleration at speed 3 m/s on a radius-2 m circle. Repeat the interpretation when an ideal guide supplies all normal reaction.
7. Let $p(\lambda) = \ell(\lambda, \sin(\pi\lambda), 0)$ for $0 \le \lambda \le 2$, where $\ell = 1$ m. Derive the curvature and the conversion from $\dot{\lambda}$ to physical speed. Formulate a timing optimization with a total-acceleration bound, a finite speed bound, and zero initial and terminal speeds; do not assume that a bound on normal acceleration alone specifies a smooth minimum-time solution.
8. Starting with a declared double-pendulum model, generate a smooth swing-like segment and compare configuration, physical clubhead, and normalized state curvature. Repeat under different state scales and sampling noise. Report which features remain stable and which do not.
9. Construct two congruent physical paths with different positions relative to a fixed ball. Explain why their curvature signatures agree while their impact outcomes differ.
10. For a measured or simulated swing, propose one geometric hypothesis, its mechanical interpretation, a competing explanation, an independent validation measurement, and a criterion that could falsify it.

Chapter 3

Configuration Manifolds

3.1 Describe the System Before Choosing Coordinates

A club has a position and an orientation. An elbow angle describes a relative configuration, while clubface orientation describes an output of the entire kinematic chain. Angular velocity adds information that a configuration does not contain. These distinctions matter when comparing swings: a small change in a joint coordinate can accompany a large change in an impact outcome, and a large coordinate jump can occur while the physical motion remains smooth. Geometry helps us distinguish these possibilities. It does not determine which motion a golfer should make without a mechanical model and an outcome criterion.

Coordinates are measurements expressed as numbers. They are legitimate local descriptions, not false versions of the physical system. A useful model states what configurations are identified, what constraints apply, and where its coordinates remain valid. Some mechanical configuration spaces are Euclidean; others require multiple charts or a constrained representation. The relevant question is whether the chosen description covers the motion being analyzed without ambiguity that the analysis fails to handle.

An ideal revolute joint with unlimited rotation has configuration space S^1 when a full turn restores every modeled property. If cable winding, accumulated twist, or another internal variable distinguishes successive turns, that angle alone is not a complete configuration. A joint restricted between two stops instead has an interval of admissible angles. Its interior admits a single real coordinate; the endpoints require a boundary or contact model. These models answer different questions and should not be interchanged silently.

A smooth n-dimensional manifold is a Hausdorff, second-countable space locally described by open subsets of $\mathbb{R}^n$, with smooth invertible transitions between overlapping coordinate charts. This definition covers $\mathbb{R}^n$ itself. A chart need not cover the entire space, but some manifolds do admit one global chart. A circle does not admit a single nonsingular real coordinate with all its identifications preserved. Embedding it as $(\cos\theta, \sin\theta)$ uses two numbers subject to one constraint rather than creating a second degree of freedom. Boundaries, corners, changing contact modes, and singular constraint sets require qualifications beyond an ordinary smooth manifold.

3.1.1 Configurations, Directions, and Task Outputs

A spherical pendulum whose only configuration is the direction of a rigid rod has space S^2. An unconstrained rigid-body orientation has space $SO(3)$, which also records rotation about the rod's axis. A two-axis universal joint consists of ordered revolute joints. With ideal unlimited independent angles, its joint configuration space is $S^1 \times S^1$, not automatically S^2. Its pointing-direction output may cover a sphere with repeated or singular representations. Joint limits, interference, or closure can restrict that space. Counting two degrees of freedom does not identify its topology.

An open ideal chain with two spherical joints and three revolute joints has the nine-dimensional configuration space

$$Q_0 = SO(3) \times SO(3) \times (S^1)^3, \qquad \dim Q_0 = 9. \tag{3.1}$$

This can be a teaching model of torso, shoulder, elbow, and wrist motion. It is not an anatomical assertion that those joints rotate without limits, nor a complete two-handed golfer. A real model must state its pelvis/ground attachment, joint axes, ranges, hand contacts, and whether shaft flexibility or grip compliance is represented. Shoulder translation and distributed spinal motion may matter for the chosen question.

Smooth holonomic constraints $\Phi(q) = 0$ with independent Jacobian rows define a local submanifold of dimension $n - \operatorname{rank} D\Phi$. Velocities then satisfy $D\Phi(q)\dot{q} = 0$. At changing rank, this dimension argument needs reassessment. Unilateral contact also has inequalities and mode transitions. A velocity constraint need not integrate to a configuration constraint. Consequently, a closed chain cannot be analyzed by adding nominal joint degrees of freedom and ignoring closure. Nor does configuration dimension give the number of independently identifiable hand forces or muscle inputs.

The clubhead position $p(q)$ and face orientation $R_c(q)$ are task maps out of this configuration space. Their derivatives can lose rank even when the configuration coordinates are regular. Conversely, singular coordinates can be replaced without changing the rank of a physical task map. Keeping these two questions separate is essential when interpreting a large wrist-angle change or a poorly conditioned inverse calculation.

3.2 Velocities, Forces, and Mechanical State

A tangent vector at q describes an infinitesimal configuration change. In coordinates it has components v^i, and a mechanical velocity is one such vector. The tangent bundle TQ collects all pairs (q, v). For a regular second-order mechanical model with no additional internal state, it is a natural state space of dimension $2n$. Muscle activation, viscoelastic memory, flexible modes, controller state, and contact mode can require an augmented state. A configuration path alone does not determine any of these quantities.

A generalized force is a covector $F \in T_q^*Q$: it acts on a virtual displacement to give virtual work, and on velocity to give power. In coordinates,

$$\delta W = F_i \, \delta q^i, \qquad P = F_i \dot{q}^i. \tag{3.2}$$

Repeated indices are summed here. A torque represented by a three-vector in a chosen orthonormal frame uses an identification between vectors and covectors; this convenience should not obscure the dual role of force. Mass and inertia provide a different identification in generalized coordinates.

Let new coordinates be $z = \psi(q)$ with invertible Jacobian $A = D\psi$. Velocity components transform as $v_z = Av_q$, whereas force components transform as $F_z = A^{-T} F_q$. Therefore $F_z^T v_z = F_q^T v_q$: physical power does not depend on the coordinate units or chart. Transforming both with A would generally destroy this equality. This is a direct diagnostic for incorrect unit scaling in a swing model.

For a positive-definite mass matrix $M(q)$, kinetic energy is

$$T(q, v) = \tfrac{1}{2} v^T M(q) v, \qquad M_z = A^{-T} M_q A^{-1}. \tag{3.3}$$

It defines an inner product on configuration tangent vectors. Redundant coordinates, massless idealizations, or uneliminated constraints can instead give a singular matrix; then a positive-definite metric must be justified on the appropriate reduced tangent space. The configuration mass matrix is not automatically a metric on the augmented position/velocity/activation state.

3.3 Rotations and Three Different Singularities

We use active rotation matrices $R \in SO(3)$ mapping body components into world components. Thus $R^T R = I$ and $\det R = 1$. Body angular velocity ω and world angular velocity ω_w satisfy

$$\dot{R} = R\widehat{\omega} = \widehat{\omega_w}R, \qquad \omega_w = R\omega, \qquad \widehat{a}\, b = a \times b. \tag{3.4}$$

Nine matrix entries describe three rotational degrees of freedom subject to constraints. A numerical integrator must preserve or appropriately restore these constraints. This representation avoids Euler-chart singularities, but it does not remove joint stops, loss of task authority, or input bounds.

For the specific ZYX convention $R = R_z(\psi)R_y(\theta)R_x(\phi)$, body angular velocity is related to roll, pitch, and yaw rates by

$$\omega = \begin{pmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{pmatrix} \begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix}, \qquad \det E = \cos\theta. \tag{3.5}$$

At $\theta = \pm\pi/2$ the rate map loses rank. Recovering arbitrary angle rates from a physical angular velocity is then impossible or nonunique, and nearby inversions can amplify measurement errors. This does not mean the rigid body loses a rotational degree of freedom. A pure pitch motion $R(t) = R_y(t)$ crosses $\pi/2$ with finite body velocity $(0, 1, 0)$; this special coordinate path remains finite even though the chart cannot represent all nearby motions regularly. Infinite torque does not follow from the chart alone: it depends on whether a controller performs an ill-conditioned inversion, differentiates discontinuities, saturates, or switches charts.

There are three distinct concerns. A *coordinate singularity* is a failure of a representation. A *mechanical singularity* may align actual gimbal axes or change the mobility of a mechanism. A *task singularity* is a rank loss in the map from feasible joint velocity to the selected output. A three-gimbal inertial platform can physically suffer gimbal alignment; rewriting its orientation with quaternions does not rebuild its hardware. Historical spacecraft incidents should not substitute for this distinction.

For biomechanics, report the rotation sequence, segment frames, calibration, and the observed distance to its singular set. A different Euler sequence or local chart can be entirely adequate over a measured swing. Quaternion or matrix methods are useful when coverage, interpolation, or integration makes them preferable. No sequence-independent claim that every golf shoulder passes near gimbal lock is justified without actual orientation data.

3.3.1 Exponential Coordinates and the Half-Turn Cut

For a rotation vector $a \in \mathbb{R}^3$, $\alpha = \|a\|$, Rodrigues' formula is

$$\exp(\widehat{a}) = I + \frac{\sin\alpha}{\alpha}\widehat{a} + \frac{1-\cos\alpha}{\alpha^2}\widehat{a}^2. \tag{3.6}$$

The coefficients have limits 1 and 1/2 at zero. Stable implementations use appropriate small-angle expansions. The principal rotation logarithm chooses an angle below π away from half-turns. At angle π, axes n and $-n$ give the same rotation with opposite rotation vectors. A single continuous principal branch cannot resolve that choice globally.

This branch ambiguity differs from a singular differential of the exponential map. The right-trivialized differential is

$$J_r(a) = I - \frac{1-\cos\alpha}{\alpha^2}\widehat{a} + \frac{\alpha-\sin\alpha}{\alpha^3}\widehat{a}^2, \qquad \det J_r = \left(\frac{2\sin(\alpha/2)}{\alpha}\right)^2. \tag{3.7}$$

Its determinant is $4/\pi^2$ at a half-turn and zero at a nonzero full turn. The exponential is locally regular at each half-turn vector even though the principal logarithm must choose between globally identified vectors. At zero the determinant limit is one. This example separates local invertibility from global uniqueness, two ideas that should remain distinct throughout trajectory analysis.

A unit quaternion is another rotation representation, with q and $-q$ representing the same orientation. The raw Euclidean difference can have norm two for identical physical orientations. A shortest angular distance is $2\arccos(|q_1^T q_2|)$ for normalized quaternions, with the dot product clipped for numerical roundoff. Sign-consistent interpolation or a sign-invariant error is needed. A raw quaternion difference does not generally reverse direction when only one input sign changes; the actual defect is dependence on an arbitrary representative. Lifting rotations to quaternions does not by itself guarantee global continuous feedback convergence or prevent unwinding.

3.4 What a Kinetic Metric Measures

The kinetic metric gives the length of a configuration curve $q(\lambda)$ as

$$L_M = \int \sqrt{q'(\lambda)^T M(q(\lambda)) q'(\lambda)}\, d\lambda = \int \sqrt{2T(q(t),\dot{q}(t))}\, dt. \tag{3.8}$$

It is invariant under regular monotone changes of the path parameter. It is not an energy expenditure. With ordinary mechanical units its length has units $\sqrt{\mathrm{kg}}\,\mathrm{m}$, whereas its time derivative has units $\sqrt{\mathrm{J}}$. Work also depends on applied forces, and metabolic cost requires a physiological model. The action $\int T\,dt$ is another quantity again and depends on timing.

For an illustrative pair of independent constant-inertia rotations with $I_1 = 12$ and $I_2 = 0.03\ \mathrm{kg\,m^2}$, equal angular displacements have metric lengths in ratio $\sqrt{I_1/I_2} = 20$. At equal angular speeds their kinetic energies have ratio 400. If the first motion takes twenty times as long with a correspondingly scaled speed profile, their instantaneous kinetic energies at matched path phase can instead be equal. The angle change alone establishes neither energy ratio nor required work. These numbers illustrate a model; they are not subject-specific torso and wrist inertia estimates. Coupled limbs also have off-diagonal mass-matrix terms.

Different metrics answer different questions. An orientation distance based on rotation angle is convenient for face alignment; a measurement covariance metric can quantify uncertainty; the kinetic metric weights generalized velocity by inertia. For the standard rotation-angle metric, the largest distance on $SO(3)$ is π. Arbitrary scaling or an anisotropic kinetic metric changes that statement. A clubface error measured in degrees should not be silently reinterpreted as a kinetic distance or a success probability.

3.5 Geodesics, Gravity, and Passive Dynamics

The Levi-Civita connection of a metric is the unique metric-compatible, torsion-free connection. Its coordinate coefficients are

$$\Gamma^i_{jk} = \tfrac{1}{2}(M^{-1})^{i\ell}(\partial_j M_{\ell k} + \partial_k M_{\ell j} - \partial_\ell M_{jk}), \qquad (\nabla_{\dot q}\dot q)^i = \ddot q^i + \Gamma^i_{jk}\dot q^j \dot q^k. \tag{3.9}$$

An affinely parameterized geodesic satisfies $\nabla_{\dot q}\dot q = 0$. It has constant metric speed and minimizes length on sufficiently short segments. A long geodesic need not be a globally shortest path. General reparameterization preserves its unparameterized image but can introduce tangential covariant acceleration. The geodesic equation requires its parameter convention as well as its metric.

Let $F = B(q)u + F_{\text{passive}} + F_{\text{external}}$ denote nonpotential generalized force covectors, and let V include the conservative gravity and elastic potentials chosen for the model. Avoid counting an elastic force in both V and F_{passive}. For an unconstrained regular simple mechanical system, the equivalent covector and vector equations are

$$M\nabla_{\dot q}\dot q = -dV + F, \qquad \nabla_{\dot q}\dot q = -\operatorname{grad}_M V + M^{-1}F, \qquad \operatorname{grad}_M V = M^{-1}dV. \tag{3.10}$$

Here dV means the column of partial derivatives when written in coordinates. The metric lowers a tangent index; its inverse raises a covector index. Multiplying the left side by M while leaving a Riemannian gradient on the right would mix the two types. In coordinates the same mechanics is $M\ddot q + C\dot q + dV = F$, where $C\dot q$ equals the metric-lowered connection term. The matrix C is not uniquely determined by that product.

Only when the total nonkinetic force vanishes does physical time parameterize a geodesic of the kinetic metric. Zero actuator command does not eliminate gravity, springs, dissipation, or external forces. An ideal pendulum with $M = m\ell^2$ has zero connection coefficient in its angular coordinate, yet under gravity it satisfies $\ddot\theta = -(g/\ell)\sin\theta$. With $\ell = 1$ metre and $\theta = \pi/6$, its acceleration is -4.905 rad/s^2. It is passive and conserves $T + V$, but is not a time-parameterized geodesic of that kinetic metric. Fixed-energy conservative paths can sometimes be related to a Jacobi metric with a different parameter; this is a separate construction, requiring $E > V$ and excluding the degeneracy at turning points.

Thus an observed follow-through cannot be called nearly geodesic merely because it looks relaxed. The test is a model-based covariant-acceleration residual, interpreted alongside estimated gravity, elasticity, contact, dissipation, and active moments. A small residual is not an observation of small muscle activation. Opposing muscles can have substantial activation and small net moment, and motion data may not identify those forces uniquely.

3.5.1 A Flat-Space Counterexample

In polar coordinates on the Euclidean plane away from the origin, the unit-mass metric is $M = \operatorname{diag}(1, r^2)$. Its nonzero coefficients include $\Gamma^r_{\theta\theta} = -r$ and $\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = 1/r$. A free particle obeys

$$\ddot{r} - r\dot{\theta}^2 = 0, \qquad \ddot{\theta} + 2\dot{r}\dot{\theta}/r = 0. \tag{3.11}$$

For the straight Cartesian motion $(x, y) = (1, t)$, take $r = \sqrt{1 + t^2}$ and $\theta = \arctan t$. Then $\dot{r} = t/r$, $\ddot{r} = 1/r^3$, $\dot{\theta} = 1/r^2$, and $\ddot{\theta} = -2t/r^4$. Both equations vanish exactly, despite nonzero ordinary coordinate accelerations and connection coefficients.

Intrinsic Riemann curvature is zero here. Connection coefficients are not themselves curvature tensors; they can be nonzero in curvilinear coordinates on flat space. Conversely, the topology of a configuration space does not specify its metric curvature. The coordinate terms represent a consistent description of inertial dynamics, not a source of energy. In a multibody system, inertial coupling can exchange energy between motions while the overall power balance remains

$$\frac{d}{dt}(T + V) = F_i \dot{q}^i \tag{3.12}$$

for time-independent M, V and the stated unconstrained model. Stationary ideal constraint reactions do no work on allowed velocities; moving constraints, friction, and impacts require their own power or energy terms. A distal speed increase must be reconciled with this balance, not attributed to work created by a coordinate connection.

3.6 Path Bending and Required Actuation

Let $q(s)$ be a regular configuration path parameterized by kinetic-metric arc length, $T_q = dq/ds$ its unit tangent, and $\nu = ds/dt > 0$ its metric speed. To avoid confusing kinetic energy with the tangent, the subscript is retained. Its covariant curvature vector is $k_g = \nabla_{T_q} T_q$, with norm $\kappa_g = \|k_g\|_M$. The acceleration decomposition is

$$\nabla_{\dot{q}}\dot{q} = \dot{\nu} T_q + \nu^2 k_g, \qquad Bu = M(\dot{\nu} T_q + \nu^2 k_g) + dV - F_{\text{passive}} - F_{\text{external}}. \tag{3.13}$$

The curvature vector is metric-orthogonal to the tangent. In the special case that actuators supply the entire force and there is no potential or other load, the transverse force covector is $\nu^2 M k_g$, whose dual norm is $\nu^2 \kappa_g$. The dual norm is $\|F\|_{M^{-1}} = \sqrt{F^T M^{-1} F}$. This is not an unqualified formula for muscle effort or Cartesian newtons.

In a constrained system add reaction covectors and enforce the constraints, or use a valid reduced model. In an underactuated system the right side must lie in the image of B, and a feasible solution must satisfy input bounds. Different columns can have different torque limits and biological costs. The scalar curvature does not establish this feasibility. A guide can provide normal reaction without actuator work; gravity can supply some acceleration while changing potential energy. The feasible timing calculation in Chapter 2 is therefore still required.

Geodesic curvature is not obtained by subtracting a scalar coordinate curvature from a scalar "Coriolis curvature." It is a norm of a covariant derivative in a specified metric. Ordinary curvature of a plotted coordinate path changes under coordinate scaling and can have different units. Neither is automatically the Cartesian curvature of the clubhead path $p(q)$. That physical path still obeys $\ddot{p} = J\ddot{q} + \dot{J}\dot{q}$. These are connected descriptions, but they answer different questions about the same motion.

3.7 Tracking Errors With Consistent Frames

A nearby configuration can be compared with a reference using a common chart, a local Riemannian logarithm, a retraction, or an embedded error with stated constraints. A Lie-group logarithm is another option when the configuration space actually has that group structure. It is not a universal subtraction operation on every manifold. The Riemannian exponential of a chosen metric also need not coincide with the Lie-group exponential; they agree in special settings such as the standard bi-invariant rotation metric.

Velocities at different configurations require a stated identification. Levi-Civita parallel transport is one choice and depends on the connection and path. A common coordinate chart or group trivialization is another. Rotating components between body frames is a physical frame transformation; it should not be casually labeled parallel transport for an arbitrary kinetic metric. These conventions must be settled before constructing feedback gains.

3.7.1 A Derived Rigid-Body Tracking Law

Consider a fully actuated rigid body with constant body inertia $J > 0$, $J\dot{\omega} + \omega \times J\omega = \tau$. Assume an exactly known model, a smooth feasible desired rotation $R_d(t)$, and available torque without saturation for the following derivation. Define $\dot{R}_d = R_d\widehat{\omega}_d$ and $A = R^T R_d$, which maps desired-body components into actual-body components. Let

$$e_\omega = \omega - A\omega_d, \qquad \Psi = \tfrac{1}{2}\operatorname{tr}(I - R_d^T R), \qquad e_R = \tfrac{1}{2}(R_d^T R - R^T R_d)^\vee. \tag{3.14}$$

The vee map inverts the hat map on skew matrices. This smooth e_R is $\sin\alpha$ times the relative rotation axis, not the logarithmic angle-axis vector αn. In particular it vanishes at a half-turn, an important limitation rather than a numerical defect to conceal.

Differentiating the frame transformation gives

$$\dot{A} = -\widehat{\omega}A + A\widehat{\omega}_d, \qquad b = \frac{d}{dt}(A\omega_d) = -\omega \times (A\omega_d) + A\dot{\omega}_d. \tag{3.15}$$

Using the inverse transformation $R_d^T R$ in e_ω would compare vectors in inconsistent frames. Omitting the first term in b would omit the rate at which the reference components rotate relative to the actual body.

For positive scalar gains, choose

$$\tau = -k_R e_R - k_\omega e_\omega + \omega \times J\omega + Jb. \tag{3.16}$$

Substitution yields $J\dot{e}_\omega = -k_R e_R - k_\omega e_\omega$. Direct differentiation of Ψ gives $\dot{\Psi} = e_R^T e_\omega$; hence

$$W = \tfrac{1}{2}e_\omega^T J e_\omega + k_R\Psi, \qquad \dot{W} = -k_\omega\|e_\omega\|^2. \tag{3.17}$$

For $W(0) < 2k_R$, the trajectory stays away from the half-turn potential level. In the ideal closed error dynamics, the largest invariant zero-dissipation set in this region has $e_\omega = 0$, then $e_R = 0$, and therefore the desired relative orientation. This supplies a local attraction argument. Stronger rate or domain claims require the corresponding proof and assumptions. At a half-turn with zero velocity error the feedback vanishes, providing an explicit counterexample to global attraction. A globally defined smooth formula is not the same as a globally convergent controller. Logarithmic feedback has its own branch restriction; quaternion feedback must respect the double cover, and hybrid designs require analysis of their switching.

Two checks make the formula practical to audit. If actual and desired bodies have the same world angular velocity w, then $\omega = R^T w$ and $\omega_d = R_d^T w$, so $e_\omega = 0$ exactly. For arbitrary orientations and rates, substituting the torque into the rigid-body equation must reproduce the stated $\dot{W}$. Both identities can be verified numerically without simulating a golf swing. Torque saturation, disturbances, uncertain inertia, underactuation, and muscle activation dynamics change this closed-loop result. It is a mathematical example, not an identified human motor-control law.

3.8 Connecting Geometry to a Golf Investigation

Begin with the observable outcome: preimpact clubhead velocity, face orientation, strike location, and their uncertainty at a defined event. Construct the kinematic model needed to map measured segment poses to those outcomes. State which joints, constraints, flexible elements, and contact modes the model retains. A larger geometric model is useful only when its parameters and outputs can be estimated well enough for the question.

Next verify representation consistency. Record active/passive rotation conventions, handedness, body/world frames, joint sequence, units, quaternion ordering, and calibration. Test that converting coordinates preserves physical orientations, velocities, kinetic energy, and power. Examine chart conditioning along the actual data. A discontinuity created by angle wrapping is not evidence of a sudden physical release, and smoothing it as if it were physical can corrupt estimated acceleration and inverse torque.

Then connect kinematics to dynamics. Estimate the forces and moments required by the motion, including gravity, elastic energy, hand and ground interactions, and the shaft model. Propagate measurement and parameter uncertainty. Compute covariant quantities only with a declared metric and compare them with physical clubhead quantities through the task Jacobian. Apparent simplicity in one coordinate system does not establish lower loading, greater efficiency, or better impact reliability.

Finally test the explanatory hypothesis. If inertial coupling is proposed to assist distal speed, verify the energy and power balance and compare forward simulations under a precisely defined change in coupling, activation, or timing. Recompute the resulting motion rather than subtracting a term from a fixed trajectory and calling it a performance prediction. Check predictions on held-out swings and report where uncertainty prevents discrimination between mechanisms. Passive assistance, active regulation, and task constraints may operate together; their contributions are questions to estimate.

A configuration tube $d_M(q, q_d(t)) \leq \varepsilon$ bounds only configuration error under its chosen metric. Two states inside it can have very different velocities and impact results. A state tube needs a metric or certificate on the actual state space, with explicit velocity comparisons and contact events. Invariance must be verified under the closed-loop dynamics and disturbances. Geometry defines the set; dynamics determine whether a trajectory stays inside, and the task map determines whether staying inside is useful.

The benefit of manifold methods is precise bookkeeping of configuration, velocity, force, and frame relationships across a motion. That precision strengthens an explanation of the swing when combined with measured evidence. It does not remove the need for timing, feasibility, stability, or an empirical account of how the golfer supplies and regulates forces.

3.9 Further Reading

Lynch and Park's Modern Robotics provides joint, rotation, wrench, and rigid-body conventions. Lee, Leok, and McClamroch's geometric control analysis develops smooth attitude tracking with explicit attraction conditions. These are mathematical and engineering sources; applying their structures to a human swing requires the additional modeling and validation steps described here.

3.10 Exercises

1. Compare an unlimited revolute joint, a joint with stops, and a rod whose direction alone is measured. Identify the configuration, output map, and any unobserved rotation. Explain why dimension alone cannot identify topology.
2. For an ideal chain with two spherical and three revolute joints, count configuration and tangent-bundle dimensions. Add two independent holonomic constraints locally. Explain why an activation state changes the state count.
3. Derive the ZYX body-rate matrix and its determinant. Evaluate a pure pitch crossing through the rate-map singular set at $\theta = \pi/2$. Distinguish loss of inverse uniqueness from unbounded physical velocity.
4. Evaluate $\det J_r$ at 0, π, and 2π. Compare the exponentials of πn and $-\pi n$. Explain why local invertibility at each half-turn does not define a globally continuous principal logarithm.
5. Choose a nonsingular coordinate scaling and verify the velocity, force, and mass-matrix transformation laws. Check kinetic energy, power, and dual force norm numerically before and after transformation.
6. Verify the polar-coordinate free-particle example and compute a component of its Riemann curvature tensor. Contrast nonzero connection coefficients with zero intrinsic curvature and zero Cartesian acceleration.
7. Derive the gravity pendulum equation and its energy balance. Compare its motion with the kinetic-metric geodesic from the same initial state. Explain what zero actuator torque does and does not imply.
8. For a regular configuration path, derive the required actuator covector including a potential and external force. State the image and bound tests needed when the actuation map is rectangular.
9. Implement the attitude law with two different body orientations but a common world angular velocity. Verify zero velocity error, the transport derivative, and the dissipation identity. Test the undesired half-turn equilibrium and a torque-saturated case separately.
10. Design a swing-data comparison that preserves the physical motion while changing its coordinates. Identify quantities that must remain invariant and quantities that depend on the chosen metric. Add a separate intervention that changes the dynamics and specify the impact prediction it would test.

Chapter 4

Orbital Stability and Transverse Linearization

4.1 Choose the Error That Matches the Task

A swing can follow a similar spatial path at a different speed, arrive at impact at a different time, or deliver a different face orientation at the same time. These are different perturbations. Their importance depends on the task and on which variables the path contains. A configuration path omits velocity; a mechanical state-space path includes it. Sliding along the latter is therefore not permission to choose an arbitrary speed along the former. The compatibility conditions from Chapter 1 still apply.

Timed tracking compares $x(t)$ with a feasible reference $x_d(t)$ at the same clock time. Orbital analysis compares a state with an invariant trajectory set. Event-based analysis compares outcomes when a specified event occurs. All three can be useful. A stationary ball may make some absolute start-time shifts irrelevant, while clubhead speed, face orientation, contact order, and timing relative to an external disturbance remain consequential. The choice should follow the outcome, not a general preference for or against time-indexed error.

Lyapunov stability means that sufficiently small initial perturbations remain small. Asymptotic stability additionally requires convergence; exponential stability specifies a rate bound. Lyapunov functions can analyze equilibria, time-varying tracking errors, invariant sets, and periodic orbits. Orbital stability is not an alternative that invalidates Lyapunov analysis. It is a choice of the set whose stability we want to study.

For the unit-speed circle $x_d(t) = (\cos t, \sin t)$, a phase-shifted solution $x_d(t - \varepsilon)$ has same-time distance

$$\|x_d(t - \varepsilon) - x_d(t)\| = 2|\sin(\varepsilon/2)|. \tag{4.1}$$

This remains small with small ε and does not decay. It disproves asymptotic convergence to the scheduled solution, not stability. For a system driven by an explicit clock-dependent input, the shifted curve is not generally a solution under the original input. Either shift the input too, use a phase-dependent autonomous feedback law, or include the forcing phase in the state before drawing an orbital conclusion.

4.2 Invariant Orbits and Finite Segments

Consider a smooth autonomous closed-loop system $\dot{x} = F(x)$ with a regular nonstationary periodic solution $\gamma(t + T) = \gamma(t)$ of minimal period T. Its image $\mathcal{O}$ is an invariant closed curve. In a neighborhood with a specified distance, orbital stability means

$$\forall \epsilon > 0 \ \exists \delta > 0 : \quad d(x_0, \mathcal{O}) < \delta \implies d(x(t), \mathcal{O}) < \epsilon \quad \text{for all } t \geq 0. \tag{4.2}$$

Asymptotic orbital stability adds $d(x(t), \mathcal{O}) \to 0$ locally. Asymptotic phase additionally means convergence to a particular time-shifted nominal solution, $\|x(t) - \gamma(t + \theta_*)\| \to 0$. These statements require the relevant existence, invariance, and neighborhood conditions; they are not inferred from a visually closed plot of a few coordinates.

A finite downswing segment is generally not invariant: its nominal solution leaves through the endpoint. It cannot satisfy an all-future-time set-stability definition merely because nearby motions remain close until impact. Useful finite-horizon questions instead concern error amplification, constraint satisfaction, disturbance sensitivity, and event outcomes before termination. They need their own bounds. Calling a finite segment an orbit for descriptive purposes should not transfer infinite-time theorems to it.

A nonstationary smooth periodic orbit is regular because its autonomous flow cannot pass through an equilibrium and then leave under uniqueness. Arbitrary reference curves can have stops, self-intersections, or endpoints. A projection into selected measurements can intersect itself even if the full state orbit does not. The state, metric, regularity, and selected phase neighborhood must therefore be specified before constructing transverse coordinates.

The Van der Pol oscillator has an attracting limit cycle for positive damping parameter in its standard form. It is a useful mathematical example. Walking is often modeled with hybrid periodic gaits, and a golf swing is usually modeled as a transient. Neither an analogy with an oscillator nor an attractive orbit in a reduced model establishes a measured human coordination strategy.

4.3 Derive the Local Phase and Transverse Coordinates

Work first in a fixed Euclidean state chart with declared scaling. Let $\gamma(\tau)$ be a regular reference, parameterized by its nominal physical time τ, and let $u_*(\tau)$ satisfy $\gamma'(\tau) = f(\gamma(\tau), u_*(\tau))$. Define $v(\tau) = \|\gamma'(\tau)\| > 0$, unit tangent $T = \gamma'/v$, and a smooth orthonormal normal basis $E(\tau) \in \mathbb{R}^{n\times(n-1)}$, with $E^T E = I$ and $E^T T = 0$. A local coordinate representation is

$$x = \gamma(\tau) + E(\tau)\xi, \qquad \xi \in \mathbb{R}^{n-1}. \tag{4.3}$$

The ambient displacement $E\xi$ has n components; the reduced error ξ has $n-1$. The orthogonal projector $P = EE^T = I - TT^T$ is an $n \times n$ matrix. Projecting with P does not itself produce reduced coordinates.

For nearest-point phase, $T(\tau)^T[x - \gamma(\tau)] = 0$. This defines a smooth local phase only where its derivative with respect to τ is nonzero and the chosen closest point is unique. A compact embedded curve has an appropriate tubular neighborhood, but a curvature radius alone does not establish its global width: distant pieces of the curve can approach one another. Endpoints and branch changes need explicit handling. Frenet frames can fail at zero curvature; a smooth normal frame can often be constructed without that failure.

Differentiate the coordinate representation and project along T and E. For $u = u_*(\tau) + \delta u$, the exact local equations are

$$\dot{\tau} = \frac{T^T f(x,u)}{v - T'^T E\xi}, \qquad \dot{\xi} = E^T f(x,u) - E^T E'\xi\dot{\tau}. \tag{4.4}$$

Primes here mean derivatives with respect to nominal time τ, and dots mean actual physical time. We used $T^T E' = -T'^T E$. The denominator must remain nonzero; forward progression additionally requires $\dot{\tau} > 0$. At the reference $\dot{\tau} = 1$ and $\dot{\xi} = 0$. This is the nominal-time version of Chapter 2's arc-length equations, with its parameter factors made explicit.

Linearization at fixed phase gives

$$\dot{\xi} = A_\perp(\tau)\xi + B_\perp(\tau)\delta u + O(\|\xi\|^2 + \|\delta u\|^2), \tag{4.5}$$

$$A_\perp = E^T f_x E - E^T E', \qquad B_\perp = E^T f_u. \tag{4.6}$$

Mixed second-order terms are absorbed into the displayed bound locally. The dimensions are $(n-1)\times(n-1)$ and $(n-1)\times m$, respectively. The frame derivative matters; an arbitrary sequence of independently computed normal bases can introduce sign jumps and fictitious dynamics. In a rotating normal basis this term transforms with the basis, preserving the physical prediction when the full transformation is retained.

The derivation assumes an autonomous plant and feedforward evaluated at the estimated phase. Explicit time dependence, an independent clock-driven input, or a dynamic phase estimator adds variables or coupling terms. For a manifold state use compatible local charts, connections, or group coordinates as in Chapter 3. The fixed-chart formula should not be applied unchanged to an arbitrary position-dependent metric. Under a different path parameter, both the speed and cost integration measure must be transformed consistently.

4.3.1 An Oscillator With a Missing Input Factor

For $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + u$, take $\gamma(\tau) = (\cos\tau, -\sin\tau)$ and $u_* = 0$. Then $T = (-\sin\tau, -\cos\tau)^T$, $E = (-\cos\tau, \sin\tau)^T$, and $x = (1-\xi)\gamma(\tau)$. The equations become

$$\dot{\xi} = u\sin\tau, \qquad \dot{\tau} = 1 - \frac{u\cos\tau}{1-\xi}, \qquad A_\perp = 0, \quad B_\perp = \sin\tau. \tag{4.7}$$

Choose $u = -k\sin\tau\,\xi$ with $k > 0$. The exact radial equation is $\dot{\xi} = -k\sin^2\tau\,\xi$; along the nominal orbit $\tau = t$ its scalar linearization has one-period multiplier

$$\mu = \exp\left(-k\int_0^{2\pi}\sin^2 t\,dt\right) = e^{-k\pi} < 1. \tag{4.8}$$

This is a local exponential-stability certificate with the usual smoothness and regular-phase assumptions. No averaging is required for the multiplier. The incorrect law $u = -k\xi$ instead gives exponent $-k\int_0^{2\pi}\sin t\,dt = 0$ and multiplier one. The missing sine factor changes the conclusion. For sufficiently small ξ, forward phase remains well-defined; this local argument does not establish global behavior at the origin, where radial phase is undefined, or under torque saturation.

The related exercise $\dot{x}_1 = x_2 + u$, $\dot{x}_2 = -x_1^3$ cannot track the same timed circle by any choice of u: the second equation would require $-\cos t = -\cos^3 t$ for all t. Checking the uncontrolled equation should precede linearization, optimization, or feedback design.

4.4 What Linearization Can Certify

For a smooth invariant orbit, uniform exponential stability of the transverse linearization implies local exponential orbital stability under standard regularity conditions. An unstable transverse multiplier gives local instability. A marginal linearization generally leaves nonlinear behavior undecided. Nonlinear asymptotic stability is not equivalent to asymptotic stability of the linearized system.

For example, in a local annulus use angular phase $\dot{\theta} = 1$ and radial error dynamics either $\dot{\xi} = -\xi^3$ or $\dot{\xi} = +\xi^3$. Both have the same circle and the same transverse linearization $\dot{\xi} = 0$. Yet

$$\xi_-(t) = \frac{\xi_0}{\sqrt{1 + 2\xi_0^2 t}}, \qquad \xi_+(t) = \frac{\xi_0}{\sqrt{1 - 2\xi_0^2 t}}. \tag{4.9}$$

The first converges algebraically; the second grows until it leaves the local region. Both linearizations have multiplier one. Higher-order terms decide the distinction. This counterexample also explains why a linear Lyapunov function is a candidate for nonlinear verification, not an automatic proof over a large tube.

4.5 Periodic Orbits, Return Maps, and Hybrid Events

For a periodic transverse linear system $\dot{\xi} = A_\perp(t)\xi$, the state transition matrix satisfies $\partial_t \Phi(t, t_0) = A_\perp(t)\Phi(t, t_0)$, $\Phi(t_0, t_0) = I$. Its one-period map $M_\perp = \Phi(T, 0)$ has Floquet multipliers as eigenvalues. All magnitudes strictly below one give exponential stability of this periodic linear system and local exponential stability of the smooth nonlinear orbit. Any magnitude above one implies instability. For the linear system, semisimple unit-modulus multipliers can permit bounded solutions; for the nonlinear orbit they are inconclusive without more analysis.

A nonstationary autonomous periodic orbit has a unit multiplier in the full variational system because its tangent is a periodic solution of that system. There may be additional unit multipliers. Transverse reduction removes the phase direction, not every possible neutral mode. Periodically forced systems do not automatically have the same neutral phase direction when the forcing clock is held fixed. An augmented phase formulation must state what is varied.

A local Poincare section crosses the flow transversely near a point on a periodic orbit. The local return map has a fixed point there. Its derivative represents the transverse monodromy map up to a consistent change of section coordinates; numerical matrices need not be literally equal under different bases. Their nontrivial multipliers agree. A phase basis used around a cycle must also be identified consistently after one period; a frame mismatch can alter an incorrectly assembled return matrix.

A hypothetical return map with diagonal multipliers $0.7, 0.4, -0.3$ contracts those eigenmodes, with the last alternating sign. Along the first eigenmode, ten returns give factor $0.7^{10} \approx 0.028$. This is an illustrative map, not a computed walking or golf result. Nonnormal eigenvectors can produce large norm growth before asymptotic decay. A gain that moves a multiplier cannot be computed from the multiplier alone: the input map is required. For $z_{j+1} = 1.1z_j + bu_j$, $u_j = -kz_j$, the target 0.6 requires $k = 0.5/b$ when $b \neq 0$.

Hybrid motion requires the event and reset derivative as well as continuous variational integration. For a smooth guard $h(x, t) = 0$, reset $x^+ = \Delta(x^-, t)$, and transverse crossing $h_t + h_x f^- \neq 0$, the saltation matrix mapping perturbations at a common nominal time is

$$S = \Delta_x + \frac{(f^+ - \Delta_x f^- - \Delta_t)h_x}{h_t + h_x f^-}. \tag{4.10}$$

This includes the effect of changed event time. A hybrid cycle's linear map composes continuous transitions and these event maps in chronological order, with consistent coordinates. Grazing, simultaneous contacts, or changes in event order can invalidate this smooth local formula.

Even an identity reset does not imply $S = I$. In one dimension, let speed change from one to two when x crosses zero. Then $S = 2$: a small position lead triggers the faster mode earlier and doubles the subsequent position lead. Differentiating only the identity reset would miss this effect. Club-ball contact, ground contact, and changing grip constraints likewise need a stated event model; a smooth downswing Jacobian alone does not describe sensitivity across an impact.

4.6 Finite-Time Amplification and Impact Sensitivity

For a transient, $\Phi(t_2, t_1)$ gives first-order sensitivity under its stated closed-loop dynamics. In a fixed Euclidean norm its largest singular value is a worst-case amplification factor. A value above one means *some* initial direction is amplified, not all directions. A terminal value below one does not exclude earlier growth. Higher-order errors and ongoing disturbances also remain outside that homogeneous linear map.

The stable constant matrix

$$A = \begin{pmatrix} -1 & 10 \\ 0 & -1 \end{pmatrix}, \qquad \Phi(t,0) = e^{-t} \begin{pmatrix} 1 & 10t \\ 0 & 1 \end{pmatrix} \tag{4.11}$$

has both eigenvalues at -1. Nevertheless $\|\Phi(1,0)\|_2 > 3.7$, while $\|\Phi(10,0)\|_2 < 0.005$. A small final error does not prove an intermediate joint or contact constraint was respected. Bounds should cover the whole relevant interval and the actual output, not just a terminal state norm.

With positive-definite endpoint metrics W_1, W_2, the consistent gain is

$$\rho_W = \|W_2^{1/2} \Phi(t_2, t_1) W_1^{-1/2}\|_2. \tag{4.12}$$

Changing coordinate units without transforming the metrics changes the numerical gain and can change whether it exceeds one. Output sensitivity may matter more than the norm of the full transverse state: some directions barely affect face orientation, while others have large task consequences. For phase-to-phase integration, convert $d\xi/dt$ to $d\xi/ds$ with the phase rate; do not substitute a path coordinate for time in an unchanged ODE.

Impact is usually an event rather than a fixed clock time. Let the nominal event occur at t_e, with $a = h_t + h_x f \neq 0$ and a pre-event fixed-time perturbation $\delta x = \Phi(t_e, t_0)\delta x_0$. To first order,

$$\delta t_e = -\frac{h_x \delta x}{a}, \qquad \delta x_e = \delta x + f\delta t_e, \qquad \delta y = H_x \delta x + (H_x f + H_t)\delta t_e. \tag{4.13}$$

Here $y = H(x_e, t_e)$ is the event output, and all derivatives are evaluated on the nominal pre-event state. A reset requires its own derivative afterward. The denominator reveals why near-grazing events can have large timing sensitivity even when fixed-time state perturbations look modest.

For constant-speed travel $\dot{q} = v$, $\dot{v} = 0$ toward $q = L$, the fixed-time transition is $\Phi = \left(\begin{smallmatrix} 1 & t_e \\ 0 & 1 \end{smallmatrix}\right)$. At the event, position perturbation is zero by definition, but velocity perturbation remains and $\delta t_e = -(\delta q_0 + t_e \delta v_0)/v$. A metric that discards event time is appropriate only if the task also discards it. A metric that discards velocity would miss a central contributor to impact performance.

4.7 LQR With the Correct Variables and Cost

For the reduced linear model, define the finite-horizon cost

$$J = \xi(T)^T S_T \xi(T) + \int_0^T (\xi^T Q(t)\xi + \delta u^T R(t)\delta u)\, dt, \qquad Q, S_T \succeq 0, \quad R \succ 0. \tag{4.14}$$

With appropriate continuous bounded coefficients and positive-definite R, the finite-horizon Riccati solution gives

$$-\dot{S} = A_\perp^T S + SA_\perp - SB_\perp R^{-1} B_\perp^T S + Q, \quad S(T) = S_T, \quad K = R^{-1} B_\perp^T S. \tag{4.15}$$

The input law is

$$u = u_*(\tau) - K(\tau)\xi, \qquad \xi = E(\tau)^T [x - \gamma(\tau)]. \tag{4.16}$$

Its correction has m components. Multiplying $K\xi$ by the state basis E is generally dimensionally invalid: that basis maps reduced state error to ambient state displacement, not to actuator input. The negative sign follows the stated gain convention. Finite-horizon optimality for this linear cost is not an unlimited-time stability theorem or a nonlinear optimum.

Evaluating a nominal-time gain schedule at measured phase creates a phase-dependent controller. The exact dynamics include $\dot{\tau}$, so an off-reference nonlinear certificate must differentiate $S(\tau)$ with that actual rate. If the optimization itself uses another phase coordinate s, then $dt = ds/\dot{s}$ changes the dynamics and running cost. A frozen nominal conversion can be a local design approximation, but should not be described as an exact nonlinear optimal-control solution.

The value $V = \xi^T S \xi$ satisfies

$$\dot{V} = -\xi^T (Q + K^T RK)\xi \tag{4.17}$$

along the ideal linear closed loop. If $S > 0$, a constant level set gives an invariant time-varying ellipsoid for that linear system. Its axis radii are $\sqrt{c/\lambda_i(S)}$, but they need not decrease toward the terminal time. For a scalar integrator with $Q = R = 1$, $T = 1$, and $S_T = 1/4$,

$$S(t) = \tanh(1 - t + \operatorname{arctanh}(1/4)). \tag{4.18}$$

This positive solution decreases toward $1/4$, so a fixed-level ellipsoid expands. A larger terminal penalty or state weight does not by itself prove monotone tightening, feasible torque, or improved impact variance. Those claims require the resulting closed-loop and task calculations.

The left panel evaluates the triangular transition matrix above; the right evaluates the scalar Riccati radius with level one. Both are calculated counterexamples, not measured golf profiles. Together they separate asymptotic stability, intermediate amplification, and the geometric width of a verified set.

Standard full-state time-varying LQR can use nonuniform weights and explicitly represent task-relevant directions. It does not inherently penalize all errors equally. Transverse design is useful when phase freedom matches the task and the phase estimate is reliable. Timed tracking is useful when schedule matters. The comparison should use equal constraints, uncertainty assumptions, and outcome criteria, rather than assume one architecture always expends less effort.

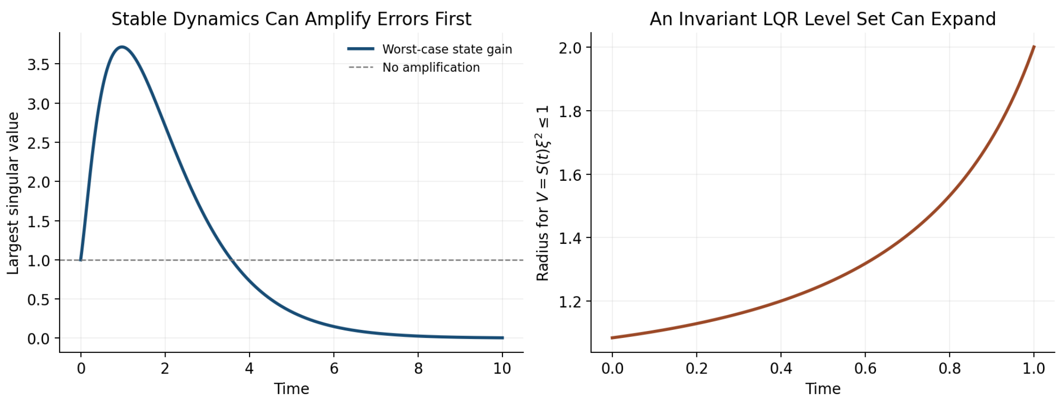


Figure 4.1: Transient Growth and Expanding Invariant Sets

4.8 Verify Nonlinear Tubes and Disturbance Effects

For a candidate tube $V(\tau, \xi) \leq c(\tau)$, verify on its boundary

$$V_\tau \dot{\tau} + V_\xi \dot{\xi} \leq c'(\tau)\dot{\tau} \tag{4.19}$$

under the actual nonlinear controller, admissible disturbances, and input bounds, with the initial condition inside. Also verify phase-coordinate validity, event ordering, and any reset inclusion into the next tube. A positive Riccati matrix is a useful starting candidate, not a replacement for these checks. For $\dot{\xi} = -\xi + \xi^3$, the quadratic $V = \xi^2$ has $\dot{V} = -2\xi^2 + 2\xi^4$: the linearized dissipation holds locally but fails outside $|\xi| = 1$. Available actuator authority alone does not repair an unverified nonlinear inequality.

For additive linear disturbances, the propagation is

$$\xi(t) = \Phi(t, 0)\xi_0 + \int_0^t \Phi(t, s)D(s)d(s)\, ds. \tag{4.20}$$

Even strong attenuation of initial error does not eliminate disturbances arriving near impact. For white-noise diffusion with covariance intensity Q_d, a linear covariance model satisfies $\dot{\Sigma} = A_{\mathrm{cl}}\Sigma + \Sigma A_{\mathrm{cl}}^T + DQ_dD^T$. This is a stochastic-model assumption, not a universal description of motor noise. Event-output covariance additionally uses the event sensitivity map, and uncertainty in estimated phase or model parameters must be included.

Classical gain and phase margins concern specified frequency-domain loop models under particular assumptions. An arbitrary time-varying transverse system has no ordinary stationary transfer function to which those margins can be assigned unchanged. Setting $R = rI$ in finite-horizon LQR does not establish universal per-channel half-gain or sixty-degree margins for a time-varying nonlinear swing controller. Appropriate alternatives include verified Lyapunov inequalities over an uncertainty set, induced-gain bounds, delay-aware models, and explicit perturbation studies. Simulation evidence should be distinguished from a worst-case certificate.

4.9 A Reproducible Golf Stability Investigation

Define the measured state and the impact event first. State whether the comparison is at common time, common phase, or each swing's own impact, and which output coordinates determine success. Include clubhead velocity and face orientation with their measurement uncertainties rather than treating proximity in an arbitrary state plot as sufficient evidence of accuracy.

Next construct a feasible nominal motion and identify what is held fixed in the perturbation experiment: prescribed torques, phase-dependent feedback, an estimated biological controller, or a passive model. A nominal trajectory does not identify its surrounding closed-loop vector field. Different feedback laws can generate the same nominal swing and different stability. Inverse dynamics supplies model-dependent net force requirements; it does not alone reveal how those forces change after a perturbation.

Compute the variational dynamics, phase transformation, and event/reset maps for that model. Test numerical derivatives against finite perturbations and check convergence with integration tolerance and perturbation size. Examine weighted singular values throughout the motion, not just eigenvalues or the final gain. Propagate realistic ongoing disturbances and inspect input, contact, and joint constraints. Estimate nonlinear validity by direct tests or a certificate with a stated region.

Only then compare a proposed passive-assistance mechanism with a changed model or controller. Recompute the trajectory and impact event under the intervention. Report whether a difference comes from initial-error attenuation, disturbance rejection, timing, output sensitivity, or altered nominal speed. These aspects can interact: higher speed can improve a performance objective while reducing correction time, and a stable state direction may have little effect on the ball. There is no universal numerical amplification profile through backswing, transition, release, and impact established by geometry alone.

Validate the predicted distinctions on held-out data and report alternative models that fit equally well. A reduction in a model's transverse norm is evidence about that model and metric. Calling it improved repeatability requires the event-output distribution; attributing it to human feedback requires additional identification or intervention evidence. Feedforward and feedback contributions should be measured under a stated decomposition, not assigned a fixed ninety-to-ten percentage.

4.10 Further Reading

Tedrake's limit-cycle notes and LQR treatment explain trajectory and orbital analysis within Lyapunov and optimal-control methods. Manchester's transverse-dynamics work develops local coordinates and nonlinear region verification. Kong and colleagues' saltation tutorial explains event-time corrections in hybrid linearization. Their assumptions must be matched to the swing model before transferring a stability conclusion.

4.11 Exercises

1. Compute the same-time distance between two phase-shifted unit circles. Distinguish bounded error, convergence to the scheduled solution, and orbital convergence. Explain what changes with a fixed periodic forcing clock.

2. Derive the nominal-time phase and reduced equations from $x = \gamma(\tau) + E(\tau)\xi$. State matrix dimensions and the denominator condition. Convert to an arc-length parameter and check the speed factors.
3. Verify the oscillator's two feedback laws and their linear multipliers. Simulate small radial perturbations and compare with $e^{-k\pi}$ after one cycle. Check phase positivity and input saturation separately.
4. Compare $\dot{\xi} = -\xi^3$ and $\dot{\xi} = +\xi^3$ around a periodic orbit. Explain why their identical unit multiplier does not settle nonlinear stability.
5. Test feasibility of the proposed circle in $\dot{x}_1 = x_2 + u$, $\dot{x}_2 = -x_1^3$. Identify the violated equation before attempting a feedback or optimization calculation.
6. Compute singular values of the triangular transition matrix throughout $0 \leq t \leq 10$. Compare transient gain with its eigenvalues, then repeat after coordinate scaling with and without transforming the metric.
7. Derive the event-time and output sensitivity for constant-speed travel. Check it by finite differences, then add an identity-reset speed switch and verify the saltation factor two.
8. Verify the scalar Riccati solution with terminal weight $1/4$ and plot the fixed-level radius. Explain how invariance and expanding geometric width can coexist. Test the nonlinear cubic remainder on a candidate tube.
9. For $z_{j+1} = 1.1z_j + bu_j$, compute gains producing multiplier 0.6 for $b = 1$ and $b = 2$. Explain why multiplier data alone cannot determine a gain.
10. Design a model-based swing experiment separating initial-error reduction, ongoing noise, phase uncertainty, and impact-output sensitivity. Specify an intervention and held-out evidence that would distinguish competing explanations.

Chapter 5

Underactuation and Passive Dynamics

In Plain Language 5.1. How One Input Moves Several Links

Moving the hands can accelerate a club even when a simplified model applies no wrist torque. The club's response depends on its orientation, existing velocity, gravity, and the acceleration of the hands. This is a useful way to study release. It does not establish that a golfer's wrists are freely hinged, that the arms must actively brake, or that the resulting motion is stable or optimal. Those are separate questions that the model and measurements must answer.

5.1 Instantaneous Authority and Motion Over Time

In independent configuration coordinates $q \in \mathbb{R}^n$, write

$$M(q)\ddot{q} + h(q, \dot{q}) = B(q)u, \qquad M = M^T \succ 0. \tag{5.1}$$

Here $h = C(q, \dot{q})\dot{q} + \nabla V(q)$ for an ideal rigid mechanical model; damping, elastic forces, and other loads must be added when retained. At fixed state the attainable acceleration set is

$$\mathcal{A}(q, \dot{q}) = -M^{-1}h + M^{-1}B\mathcal{U}, \tag{5.2}$$

where $\mathcal{U}$ is the admissible input set. With unrestricted inputs its affine dimension is $r = \operatorname{rank} B$. Underactuation means $r < n$. When $m < n$, an $n \times m$ matrix can have full column rank m while lacking full row rank. Multiple muscles do not guarantee independent joint torques: moment arms, force signs, activation, and force limits constrain the attainable set.

Instantaneous authority is different from reachability over time. Coupled dynamics can move an unactuated coordinate. A full-row-rank model also cannot follow every trajectory when contact, torque, power, speed, or state constraints are imposed. A walking foot has no moment at an ideal point contact, whereas distributed foot contact can support a bounded moment. The contact model matters.

5.2 A Coordinate-Consistent Double Pendulum

Let θ_1 and θ_2 be absolute angles of the arm and club measured from downward vertical in one inertial plane. Let τ_s be shoulder torque and τ_w wrist torque exerted on the club. Virtual work gives

$$\delta W = \tau_s \delta\theta_1 + \tau_w(\delta\theta_2 - \delta\theta_1), \qquad Q_\theta = (\tau_s - \tau_w, \tau_w)^T. \tag{5.3}$$

The corresponding actuator power is $\tau_s\dot\theta_1 + \tau_w(\dot\theta_2 - \dot\theta_1)$. For relative coordinates $q = (\theta_1, \theta_2 - \theta_1)^T$, let

$$\theta = Tq, \qquad T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \tag{5.4}$$

Then $Q_q = T^T Q_\theta = (\tau_s, \tau_w)^T$ and $M_q = T^T M_\theta T$. For this constant transformation, $h_q = T^T h_\theta$. The shoulder-only map $(1,0)^T$ is valid in both conventions when $\tau_w = 0$; the mass and bias terms must still use the chosen coordinates. Wrist power uses relative angular speed.

For a transparent worked model, use massless links of lengths l_1, l_2, point masses m_1 at the wrist and m_2 at the club tip, and positive gravity magnitude g. Set $\Delta = \theta_1 - \theta_2$ and $b = m_2 l_1 l_2$. In absolute angles,

$$M_\theta = \begin{pmatrix} (m_1+m_2)l_1^2 & b\cos\Delta \\ b\cos\Delta & m_2 l_2^2 \end{pmatrix}, \tag{5.5}$$

$$h_\theta = \begin{pmatrix} b\sin\Delta\,\dot\theta_2^2 + (m_1+m_2)gl_1\sin\theta_1 \\ -b\sin\Delta\,\dot\theta_1^2 + m_2 g l_2 \sin\theta_2 \end{pmatrix}. \tag{5.6}$$

These equations follow from $L = \frac{1}{2}\dot\theta^T M_\theta \dot\theta - V$ with $V = -(m_1+m_2)gl_1\cos\theta_1 - m_2 g l_2 \cos\theta_2$. They omit distributed limb inertia, a moving shoulder, shaft bending, and three-dimensional grip mechanics. Their purpose is an inspectable mechanism.

5.2.1 What Arm Deceleration Does and Does Not Imply

With zero wrist torque, the second row is

$$M_{21}\ddot\theta_1 + M_{22}\ddot\theta_2 + h_2 = 0, \qquad \ddot\theta_2 = -\frac{M_{21}\ddot\theta_1 + h_2}{M_{22}}. \tag{5.7}$$

The positive denominator does not determine the sign of this expression. $M_{21} = b\cos\Delta$ changes sign with geometry, and the velocity and gravity terms remain. With $M_{22} = 1$, $h_2 = 0$, and $\ddot\theta_1 = -2$, $M_{21} = 0.4$ gives $\ddot\theta_2 = 0.8$ whereas $M_{21} = -0.4$ gives -0.8. Both matrices with $M_{11} = 2$ are positive definite. These are a dimensionally consistent rotational-inertia example, not measured golfer parameters.

Arm deceleration can therefore accompany club acceleration in suitable states. It is not a necessary instruction to actively brake the arms: reaction torques can decelerate an arm while shoulder torque remains positive. Angular momentum of the arm-plus-club system also changes under external shoulder torque and gravity. Calling this an optimal momentum transfer requires an objective, admissible inputs, and an optimization result that these equations alone do not supply.

For the ideal conservative model,

$$\frac{d}{dt}(T_{\text{kinetic}} + V) = \tau_s\dot\theta_1 + \tau_w(\dot\theta_2 - \dot\theta_1). \tag{5.8}$$

Zero wrist actuator power does not mean zero energy flow into the club. Forces at the moving wrist can do work on the club while doing opposite work on the arm. Whole-system and segment energy budgets answer different questions; include translational boundary power and avoid counting internal power twice.

5.3 The Annihilator Test for Feasible Trajectories

On a constant-rank coordinate patch, let $N_B(q) \in \mathbb{R}^{(n-r)\times n}$ have rows spanning the left nullspace of B. Thus $N_B B = 0$. A twice differentiable configuration path with its stated timing requires the load

$$b_*(t) = M(q_*)\ddot{q}_* + h(q_*, \dot{q}_*). \tag{5.9}$$

The condition

$$N_B(q_*)b_* = 0 \tag{5.10}$$

is equivalent to $b_* \in \operatorname{range} B$: some unrestricted algebraic input produces that load. One solution is $u_* = B^+ b_*$, with possible additions from $\ker B$. The required input must also belong to $\mathcal{U}$. For example, $B = (1, 0)^T$ and $b_* = (10, 0)^T$ pass the annihilator test but fail an input bound $|u| \leq 1$. Input-rate and actuator-state constraints require additional dynamical checks. A locally smooth constant-rank pseudoinverse does not remove these restrictions.

This is a constraint on position, velocity, and acceleration along a trajectory, not generally a fixed subspace of state space. Changing the timing changes $\dot{q}_*$ and $\ddot{q}_*$, so the same geometric path may pass or fail. Contact reactions must first be eliminated consistently or retained as additional unknowns with their friction and unilateral-contact conditions.

5.4 Partial Feedback Linearization and Zero Dynamics

For an ideal collocated partition with $B = (I, 0)^T$, write

$$M_{aa}\ddot{q}_a + M_{au}\ddot{q}_u + h_a = u, \tag{5.11}$$

$$M_{ua}\ddot{q}_a + M_{uu}\ddot{q}_u + h_u = 0. \tag{5.12}$$

Choosing the acceleration command v_a requires

$$u = (M_{aa} - M_{au}M_{uu}^{-1}M_{ua})v_a + h_a - M_{au}M_{uu}^{-1}h_u. \tag{5.13}$$

The Schur complement is positive definite when $M \succ 0$. Exact cancellation assumes an accurate model and an admissible computed input. Saturation and uncertain loads invalidate the ideal equality $\ddot{q}_a = v_a$.

For output $e_a = q_a - q_a^*(t)$, zero dynamics are the remaining dynamics on $e_a = \dot{e}_a = 0$ under the input that keeps this set invariant. They depend on the output and its reference, and are generally time varying for tracking. They are not identical to the motion obtained by switching all actuators off.

No phase of a generic swing is automatically attracting. Integrate a specified nominal solution and the variational equation

$$\delta\dot{z} = A_z(t)\delta z, \qquad \dot{\Phi}_z = A_z(t)\Phi_z, \quad \Phi_z(t_0) = I. \tag{5.14}$$

Specify the output, comparison time or event, and the metric used to scale angle and velocity errors. Singular values of the appropriately scaled transition map measure finite-time amplification. A time-varying positive metric $W(t)$ satisfying

$$\dot{W} + A_z^T W + W A_z \preceq -2\lambda W \tag{5.15}$$

certifies differential decay on the interval where it holds, with metric bounds needed to translate it into a coordinate norm. Frozen-time eigenvalues and large club speed alone do not supply this certificate. A verified invariant funnel additionally checks its moving boundary, input limits, and disturbances.

5.5 Accessibility Is Not Forward-Time Reachability

For driftless systems $\dot{x} = \sum_i g_i(x)u_i$ with reversible inputs, a bracket-generating distribution on a connected manifold gives the horizontal connectivity described by Chow–Rashevskii. This does not immediately extend to a drifting mechanical system $\dot{x} = f(x) + \sum_i g_i(x)u_i$ with bounded inputs and a fixed downswing duration.

An orbit generated by vector-field flows permits both positive and negative flow times. Forward reachable sets do not reverse an unavoidable drift. For $\dot{x} = 1$, $\dot{y} = u$, $|u| \leq 1$, the fields span the plane, yet from the origin $x(T) = T$ and $|y(T)| \leq T$. No negative x is reachable for $T \geq 0$. Lie-algebra rank can support local accessibility under appropriate regularity and input assumptions; it does not certify global or small-time local controllability. Smooth orbit theorems also require care beyond a pointwise Lie-bracket rank test.

Abnormal extremals have a zero cost multiplier in Pontryagin's principle; singular arcs require separate analysis of the Hamiltonian's input dependence. Neither is diagnosed by visually rapid wrist release. Establish them for a specified optimal-control problem before assigning them a physiological meaning.

5.6 Connecting the Reduced Model to a Real Swing

The useful hypothesis is that proximal motion and earlier work prepare a state from which mechanical coupling contributes to club delivery. Testing it requires measured wrist and hand wrenches, consistent angular coordinates, a moving-base model, and uncertainty estimates. A passive hinge is one comparison model. Compare it with bounded wrist torques and compliant grips on the same task; then assess speed, orientation, dispersion, and effort separately.

Shaft deformation stores energy and changes the contact-point velocity. Muscle activation affects force capacity and impedance. Feedback changes responses to perturbations, while preparation changes the nominal trajectory itself. These mechanisms can coexist. A zero-torque comparison does not identify their unique biological contributions, and a simulated advantage must survive measurement and model-sensitivity checks before becoming a coaching recommendation.

5.7 Exercises

1. Verify the mass and bias terms from the Lagrangian and check the power identity with nonzero wrist torque in both coordinate systems.

2. Reproduce the coupling-sign counterexample. Include gravity and nonzero angular rates and identify when club acceleration changes sign.

3. Construct N_B for a rank-deficient actuator map. Give a load that passes the range test but violates an actuator bound.

4. For $\ddot{q}_u + cq_u = \sin t$, subtract a particular solution. Classify homogeneous perturbations for $c > 0$, $c = 0$, and $c < 0$; examine resonance at $c = 1$. Explain why bounded perturbations do not guarantee bounded forced motion.

5. Specify an experiment that distinguishes passive-hinge prediction error from measurement noise, wrist actuation, and shaft compliance.

Chapter 6

Trajectory Optimization

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

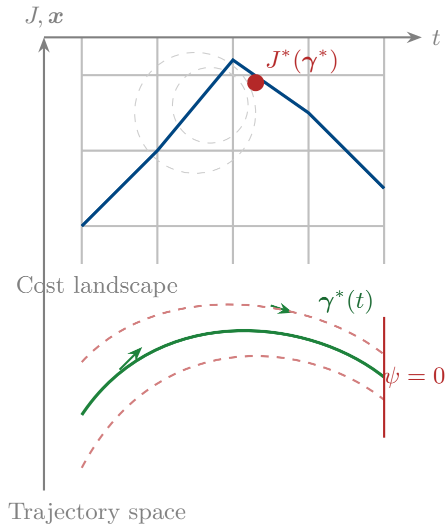
We do not merely seek a path that obeys the laws of physics.
We seek the path that bends them to our advantage.

In Plain Language 6.1. Discovering the Best Motion

How does a model propose a motion when the final task is specified but the path is not? Up until now, we have talked about how to follow a path; this chapter asks where a candidate path comes from. A trajectory optimizer can search over many admissible motions for a declared objective, such as maximizing clubhead speed at a terminal surface, while enforcing the equations of motion and actuator limits. The output is not a perfect human swing. It is a model-conditioned candidate trajectory whose usefulness depends on the cost function, constraints, parameter values, and validation data.

6.1 The Moving Target Problem

In Chapter 1, we fundamentally reframed our control objective from "reaching a destination" to "tracking a trajectory." Up to this point, we have assumed that this optimal reference trajectory $\boldsymbol{\gamma}^*(t)$ is given to us by an oracle. This chapter answers the obvious next question: *Where does this reference come from?*



For systems whose purpose is to move—like our running example of the golf swing—the optimal trajectory is not arbitrary. We are not just trying to go from point A to point B while minimizing energy. We are trying to fulfill what we term the **Moving Target Objective**: optimizing the kinematics (velocity, pose, acceleration) of a system precisely at the moment it intersects a target manifold (e.g., the golf ball).

Crucially, *the exact clock time that impact occurs does not matter.* A golfer has the freedom to start their backswing slowly and pause at the top. The optimization is entirely concerned with the *shape* of the motion and its terminal kinematics. Time is a resource, not a constraint.

Definition 6.1. The Moving Target Optimal Control Problem (OCP)

Find the state trajectory $\boldsymbol{x}(t)$, control input $\boldsymbol{u}(t)$, and terminal time t_f that minimize the cost functional:

$$J = \phi(\boldsymbol{x}(t_f)) + \int_0^{t_f} L(\boldsymbol{x}(t), \boldsymbol{u}(t))\,\mathrm{d}t \tag{6.1}$$

subject to the dynamics and constraints:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{u}(t)) \quad \text{(Dynamics)} \tag{6.2}$$

$$\boldsymbol{h}(\boldsymbol{x}(t), \boldsymbol{u}(t)) \leq \boldsymbol{0} \quad \text{(Path constraints)} \tag{6.3}$$

$$\psi(\boldsymbol{x}(t_f)) = \boldsymbol{0} \quad \text{(Terminal manifold/Moving goal)} \tag{6.4}$$

$$\boldsymbol{x}(0) = \boldsymbol{x}_0 \quad \text{(Initial state)} \tag{6.5}$$

In the golf downswing, the terminal cost $\phi(\boldsymbol{x}(t_f))$ might be negative clubhead speed (to maximize speed), the integral cost L might penalize excessive joint torques to ensure repeatability, and the terminal manifold constraint $\psi(\boldsymbol{x}(t_f)) = \boldsymbol{0}$ forces the clubhead to be geometrically located at the ball with a squared face. Notice that t_f is a *free parameter* in the optimization.

6.2 Transcribing the Continuous Problem

The OCP described above occurs in infinite-dimensional function space. To solve it on a computer, we must transcribe it into a finite-dimensional Nonlinear Programming problem (NLP):

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & f(\boldsymbol{w}) \\ \text{subject to} \quad & \boldsymbol{c}(\boldsymbol{w}) \leq \boldsymbol{0}, \end{aligned}$$

where $\boldsymbol{w}$ is a large vector of decision variables. How we discretize the integrals and differential equations defines the transcription method. We will focus on two modern approaches: **Direct Shooting** and **Direct Collocation**.

6.2.1 Direct Multi-Shooting

In direct multi-shooting, we break the time interval $[0, t_f]$ into N segments. The decision variables are the control parameters $\boldsymbol{u}_k$ on each interval and the state values $\boldsymbol{x}_k$ at the beginning of each interval.

The dynamic constraint $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ is enforced by numerical integration (e.g., Runge-Kutta). We require that the integrated state at the end of interval k exactly matches the assumed starting state of interval $k+1$:

$$\boldsymbol{x}_{k+1} - \boldsymbol{\Phi}(\boldsymbol{x}_k, \boldsymbol{u}_k, \Delta t) = \boldsymbol{0} \qquad \text{(Defect constraint)} \tag{6.6}$$

where $\boldsymbol{\Phi}$ represents the numerical integrator step. Multi-shooting is excellent for highly nonlinear dynamics because it keeps the integration intervals short, preventing the massive nonlinear sensitivities that plague single-shooting methods.

6.2.2 Direct Collocation

Direct Collocation [2] takes this a step further. We represent both the state trajectory and the control signal as piecewise polynomials (e.g., cubic splines for state, linear splines for control). The decision variables are the coefficients of these polynomials, typically located at discrete **collocation points**.

Instead of using a numerical integrator, we enforce the differential equation algebraically at the collocation points:

$$\dot{\tilde{\boldsymbol{x}}}(t_c) - \boldsymbol{f}(\tilde{\boldsymbol{x}}(t_c), \tilde{\boldsymbol{u}}(t_c)) = \boldsymbol{0} \qquad \text{(Collocation constraint)} \tag{6.7}$$

where $\tilde{\boldsymbol{x}}$ and $\tilde{\boldsymbol{u}}$ are the polynomial approximations.

Remark 6.1. Pontryagin Maximum Principle and Optimal Control

While direct collocation provides a practical numerical algorithm, the theoretical foundation for trajectory optimization comes from the **Pontryagin Maximum Principle** [1]. This principle states that along an optimal trajectory, a Hamiltonian function $\mathcal{H}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\lambda}) = L(\boldsymbol{x}, \boldsymbol{u}) + \boldsymbol{\lambda}^\top \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ must be maximized with respect to $\boldsymbol{u}$ at each instant, where the costate $\boldsymbol{\lambda}(t)$ evolves backward according to adjoint dynamics. For underactuated systems, **bang-bang** structures can emerge when the

Hamiltonian is linear in $\boldsymbol{u}$, and **singular arcs** can occur when higher-order conditions are needed to determine $\boldsymbol{u}$. A simplified golf-release model may show a control structure resembling a singular arc, but that is a hypothesis about the chosen model and bounds, not a direct prediction of measured wrist muscle activation.

where $\tilde{\boldsymbol{x}}$ and $\tilde{\boldsymbol{u}}$ are the polynomial approximations.

Key Idea 6.1. Fidelity vs. Flexibility

Direct collocation results in a massive, highly sparse NLP. Because the solver has access to the state at every grid point simultaneously, it can easily manipulate the physical trajectory to satisfy difficult path constraints without waiting to "simulate" the result. Collocation is widely considered the state-of-the-art for generating complex dynamic motions like walking, jumping, and swinging.

6.2.3 Implementation Sketch: Moving-Target Direct Collocation

For readers who want a concrete recipe, the moving-target OCP can be translated into a collocation program with only a few repeating ingredients: mesh nodes, defect constraints, terminal manifold constraints, and a continuation strategy for difficult initial guesses.

6.3 Optimizing Underactuated Geometries

Recall from Chapter 5 that the underactuated dynamics enforce a strict algebraic constraint on the system:

$$\boldsymbol{B}^{\perp}(\boldsymbol{x})\Big[\boldsymbol{M}(\boldsymbol{x})\ddot{\boldsymbol{x}} + \boldsymbol{C}(\boldsymbol{x},\dot{\boldsymbol{x}})\dot{\boldsymbol{x}} + \boldsymbol{g}(\boldsymbol{x})\Big] = \mathbf{0}. \tag{6.8}$$

When optimizing for a golf swing, direct collocation handles this underactuation organically. We do not need to explicitly partition the variables into actuated and unactuated sets for the solver. The collocation constraint $\dot{\boldsymbol{x}} - \boldsymbol{f} = \mathbf{0}$ already implicitly contains the passive dynamic constraints.

Example 6.1. The Golf Swing OCP

Consider our planar double pendulum model from Chapter 5. The solver is given the task to maximize $\dot{\theta}_2(t_f)$ (club speed).
Initial Guess: If we seed the NLP solver with a naive guess (e.g., constant shoulder torque), the club speed will be pitifully low.
The Optimized Result: The solver discovers the optimal physical orchestration entirely on its own. It will apply a massive positive u_1 to accelerate the arms, and then immediately reverse u_1 to negative maximum just prior to t_f. The optimizer mathematicaly "discovers" the necessary inertial coupling (the parametric excitation of $M_{21}\ddot{\theta}_1$) to whip the unactuated club link through the terminal manifold constraint. The passive dynamics are fully exploited to satisfy the objective.

Require: Initial state $\boldsymbol{x}_0$, dynamics $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$, running cost L, terminal manifold $\psi(\boldsymbol{x}(t_f)) = \mathbf{0}$, mesh size N.

Ensure: A feasible trajectory $\boldsymbol{\gamma}^*(t)$ and control schedule $\boldsymbol{u}^*(t)$.

Choose a normalized mesh $\tau_k \in [0, 1]$ and treat t_f as a decision variable.
Parameterize $\tilde{\boldsymbol{x}}(\tau)$ and $\tilde{\boldsymbol{u}}(\tau)$ by spline or polynomial coefficients.
Initialize the decision vector with a dynamically plausible warm start.
for each mesh interval $k = 0, 1, \ldots, N - 1$ **do**
Add collocation defects enforcing $t_f \frac{\mathrm{d}\tilde{\boldsymbol{x}}}{\mathrm{d}\tau} - \boldsymbol{f}(\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{u}}) = \mathbf{0}$.
Add path constraints $\boldsymbol{h}(\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{u}}) \leq \mathbf{0}$ at nodes or quadrature points.
end for
Add the terminal manifold constraint $\psi(\tilde{\boldsymbol{x}}(1)) = \mathbf{0}$.
Approximate the objective with quadrature:
$J \approx \phi(\tilde{\boldsymbol{x}}(1)) + t_f \sum_k w_k L(\tilde{\boldsymbol{x}}(\tau_k), \tilde{\boldsymbol{u}}(\tau_k))$.
while the NLP residuals or defect norms remain too large **do**
Solve the sparse NLP.
If convergence stalls, refine the mesh or relax the terminal target and continue homotopically.
end while
Reintegrate the optimized control on the full nonlinear dynamics to verify that the defects predict a true trajectory.
return The verified trajectory, control history, and terminal time.

Algorithm 6.1: Moving-Target Direct Collocation Workflow

6.4 Repeatability: The Stochastic OCP

The previous sections frame a deterministic problem: "Find the swing that maximizes terminal speed."

But as we argued in Chapter 1, biological systems and athletes do not optimize for peak deterministic performance. If you instruct an NLP solver to maximize golf club speed without regard for motor noise, the solver will ride the very edge of the constraint boundaries. The resulting trajectory may be extremely sensitive to execution errors; small perturbations in the state can lead to large deviations in the terminal condition $\psi(\boldsymbol{x}(t_f))$. This is quantified by the condition number of the Jacobian $\nabla_{\boldsymbol{x}}\psi$ at the optimal trajectory.

Principle 6.1. The Repeatability Addendum

The theoretically optimal trajectory is useless if its local topology amplifies small execution errors into catastrophic terminal misses. The true objective must account for *variance*.

We modify our deterministic cost $J = \phi(\boldsymbol{x}(t_f)) + \int L$ into a stochastic expectation:

$$\min_{\boldsymbol{\gamma}^*} \quad \mathbb{E}\left[\phi(\boldsymbol{x}(t_f, \boldsymbol{w}))\right] + \lambda \cdot \mathrm{Var}\left[\psi(\boldsymbol{x}(t_f, \boldsymbol{w}))\right] \tag{6.9}$$

where $\boldsymbol{w} \sim \mathcal{N}(0, \Sigma_w)$ represents signal-dependent motor noise (a known property of human muscle actuation, where noise scales proportionally to the applied torque).

To solve this in a collocation framework without massive Monte Carlo simulations, we use **Covariance Steering** or **Linear Covariance Analysis**. We augment the state vector $\boldsymbol{x}(t)$ with its local covariance matrix $\boldsymbol{\Sigma}(t)$. The dynamics of $\boldsymbol{\Sigma}(t)$ are propagated alongside the nominal trajectory using the linearized Jacobian matrices along the path.

The optimizer is now forced to "widen the funnel" as it approaches the moving goal. It sacrifices a small amount of peak clubhead speed to select a trajectory geometry where the passive dynamics are heavily stabilizing in the transverse direction. The result is a trajectory that a human can actually *repeat*.

6.5 Evolutionary Methods as Complementary Optimizers

Gradient-based transcription methods remain the primary tools of this book because they exploit analytic structure, expose multipliers and costates, and integrate naturally with the Riccati and funnel machinery developed elsewhere. Even so, there are important regimes where **evolutionary** or **genetic** methods are valuable companions rather than competitors.

When the control policy is low-dimensional, the cost landscape is badly nonconvex, or contact-rich simulation makes derivatives unreliable, practitioners often turn to genetic algorithms, evolution strategies, differential evolution, or cross-entropy style search [9, 22, 10]. These methods optimize a population of candidate trajectories or policy parameters by repeatedly scoring, selecting, mutating, and recombining them.

Remark 6.2. Where Evolutionary Search Fits

In the Control Is Motion framework, evolutionary methods are best viewed as *outer-loop search tools*. They are useful for generating warm starts, choosing high-level swing style parameters, tuning phase transitions, or exploring discontinuous task families where gradient information is poor. Once a promising family of motions has been discovered, direct collocation, DDP, and funnel verification should still be used to refine and certify the motion.

Require: Policy or trajectory parameterization $\boldsymbol{\theta}$, simulator, population size M, elite count M_e.

Ensure: A high-performing parameter vector $\boldsymbol{\theta}^\star$ that can seed a local optimizer.

1: Sample an initial population $\{\boldsymbol{\theta}^{(m)}\}_{m=1}^{M}$ around a biomechanically plausible nominal swing.

2: **for** generation $g = 1, 2, \ldots, G$ **do**

3: **for** each candidate $\boldsymbol{\theta}^{(m)}$ **do**

4: Simulate the full swing and score terminal speed, impact feasibility, and repeatability.

5: Reject candidates that violate hard constraints or miss the terminal manifold.

6: **end for**

7: Rank the feasible population by objective value.

8: Keep the top M_e elites.

9: Generate the next population by mutation, recombination, or covariance adaptation around the elites.

10: **end for**

11: Select the best feasible candidate $\boldsymbol{\theta}^\star$.

12: Use $\boldsymbol{\theta}^\star$ to initialize direct collocation or DDP for high-accuracy refinement.

13: **return** $\boldsymbol{\theta}^\star$

Algorithm 6.2: Population-Based Search for a Swing Parameterization

Key Idea 6.2. Global Search, Local Certification

A genetic or evolutionary method can help *find* a promising family of motions, but it does not by itself explain why the motion is stable or repeatable. The book's geometric pipeline remains: search globally if needed, then refine locally with derivatives, and finally certify robustness with funnels or covariance analysis.

6.6 Summary

Concept	Significance
Moving Target OCP	Finding a trajectory that optimizes terminal kinematics when intersecting a spatial manifold, without prescribing the clock time.
Multi-Shooting	Discretizing the time domain into segments, using numerical integrators to enforce dynamics constraints between nodes.
Direct Collocation	Representing the entire trajectory as splines and enforcing dynamics algebraically at discrete points arrayed along the trajectory.
Passive Exploitation	Solvers can exploit underactuated topologies through coupling matrices such as M_{21} and C_2, subject to the stated model.
Stochastic Robustness	Augmenting the OCP with expected variance to ensure the resulting trajectory can be consistently executed by noisy nonlinear plants.
Evolutionary Search	A derivative-free outer loop for discovering warm starts or task-family parameters before derivative-based refinement.

Determining the shape of the reference trajectory is central to Control Is Motion. With a passive geometry model and a candidate reference trajectory in hand, the next step is to estimate a control tube around that trajectory and state the assumptions under which disturbances remain inside it. We turn to funnel synthesis in Chapter 7.

6.7 Exercises

6.1. **(Free Terminal Time Formulation.)** Explain how one sets up an OCP where final time t_f is a decision variable in a fixed-grid Direct Collocation solver. (Hint: consider scaling time $t = \tau \cdot t_f$, where $\tau \in [0, 1]$). Show how the dynamic constraint $\dot{\boldsymbol{x}} = \boldsymbol{f}$ changes under this substitution.

6.2. **(Kinematic Goals vs Setpoints.)** You are optimizing a baseball bat swing. Your state is $(\theta, \dot{\theta})$. Assume the ball arrives at the plate between $t = 0.5$ and $t = 0.6$ seconds. Formulate the path constraint and the terminal manifold constraint ψ assuming you want maximum $\dot{\theta}$ when $\theta = \pi/2$, but the exact time of impact within that window does not matter.

6.3. **(Signal-Dependent Noise Penalty.)** Human muscle torque output τ has variance $\sigma^2 = k|\tau|^2$. Write a modified integral cost function $L(\boldsymbol{x}, \tau)$ that penalizes the introduction of variance into the system. Interpret what this does to the optimal trajectory shape.

The useful motion is rarely just the fastest.
It is the one that survives the stated disturbances.

Chapter 7

Funnel Synthesis

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

You cannot control the exact trajectory a stone will take down a mountain.
But you can build a trough to ensure it reaches the bottom.

In Plain Language 7.1. Building a Mathematical Trough

Even if a model supplies a useful reference path for a golf swing, no human executes the same path exactly twice. Muscles twitch, timing fluctuates, and wind blows. A funnel or tube around the path is a way to state a local certificate: for the declared model, controller, disturbance set, and time interval, states inside the certified set remain admissible and move toward the terminal condition. The certificate is mathematical, but it is only as strong as the model and assumptions used to compute it.

7.1 Beyond LQR: The Need for Certified Invariance

In Chapter 4, we showed that applying time-varying Linear Quadratic Regulation (LQR) to the transverse dynamics creates a locally contracting envelope of stability. The Riccati equation naturally shapes this envelope into a "funnel" that narrows as the terminal constraint approaches. The idea of composing such funnels into motion planning trees was introduced by Tedrake [23] (LQR-Trees) and extended by Majumdar and Tedrake [16] using sum-of-squares verification.

However, LQR guarantees are inherently *local*. They rely on the linear approximation of the transverse dynamics $\dot{\boldsymbol{\xi}} = \boldsymbol{A}_\perp(t)\boldsymbol{\xi} + \boldsymbol{B}_\perp(t)\delta\boldsymbol{u}$. For highly nonlinear physical systems, a disturbance large enough to leave the local neighborhood can invalidate the linear approximation. The consequence may be large terminal error or loss of the target constraint, but the threshold depends on the reference trajectory, controller, and nonlinear model.

We need a mathematically rigorous way to define a *verified boundary* for the funnel. For the fully nonlinear equations of motion, the goal is to certify that any state starting inside the candidate set at time $t = 0$ remains inside it for the modeled interval $t \in [0, t_f]$.

Definition 7.1. Dynamic Funnel

Given a reference trajectory $\boldsymbol{\gamma}^*(t)$, a feedback controller $\boldsymbol{u} = \boldsymbol{k}(\boldsymbol{x}, t)$, and a scalar boundary function $V(\boldsymbol{x}, t)$, the set:

$$\mathcal{F}(t) = \{\boldsymbol{x} \in \mathcal{X} \mid V(\boldsymbol{x}, t) \leq 1\} \tag{7.1}$$

constitutes a valid **dynamic funnel** if the set is positively invariant under the closed-loop dynamics. That is, if $\boldsymbol{x}(0) \in \mathcal{F}(0)$, then $\boldsymbol{x}(t) \in \mathcal{F}(t)$ for all $t \in [0, t_f]$.

Remark 7.1. Funnels, Reachable Sets, and What is Actually Certified

Attainable-set language from geometric control [1] is useful here, but the objects should not be conflated. A computed funnel is a verified subset of states for a chosen closed-loop model, controller, and time horizon. It may sit inside the true reachable set, coincide with only part of it, or become conservative because of the Lyapunov ansatz and SOS relaxation. Hamiltonian or symplectic structure can inform the underlying optimal-control problem, yet the numerical certificate still stands or falls on the stated closed-loop dynamics and the boundary condition $\dot{V} < 0$ on $V = 1$.

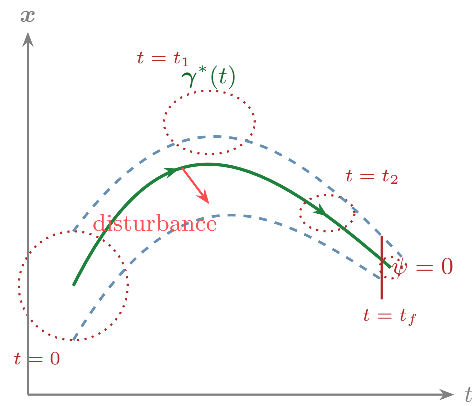
7.2 Time-Varying Lyapunov Functions for Trajectories

To verify invariance, we use the machinery of Lyapunov theory, adapted for trajectories rather than equilibria. We require our boundary function $V(\boldsymbol{x}, t)$ to act as a time-varying Lyapunov function.

Theorem 7.1 (Funnel Invariance)**.** *The set $\mathcal{F}(t) = \{\boldsymbol{x} \mid V(\boldsymbol{x}, t) \leq 1\}$ is invariant if, on the boundary where $V(\boldsymbol{x}, t) = 1$, the function is strictly decreasing along the system trajectories:*

$$\dot{V}(\boldsymbol{x}, t) = \frac{\partial V}{\partial t} + \nabla_{\boldsymbol{x}} V(\boldsymbol{x}, t) \cdot \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{k}(\boldsymbol{x}, t)) < 0 \quad \textit{for all } \boldsymbol{x} \textit{ s.t. } V(\boldsymbol{x}, t) = 1. \tag{7.2}$$

If the derivative of the boundary function points inward everywhere on the boundary, trajectories may circulate inside the funnel, but they can never escape it. For a moving target problem, we want the spatial volume of $\mathcal{F}(t)$ to shrink as $t \to t_f$, terminating in a tight bounded region at impact.



Key Idea 7.1. Funnels Define the Usable State Space

A computed funnel $\mathcal{F}(t)$ is an analysis certificate for a stated closed-loop model. It defines a conservative set of initial conditions (for example, the kinematic state at the top of the backswing) that should remain inside the verified tube and satisfy the terminal constraint $\psi(\boldsymbol{x}(t_f)) = 0$ under the modeled disturbances. The time-varying geometry of the funnel—narrowing as $t \to t_f$—encodes the stability structure of the transverse dynamics along the reference trajectory.

7.3 Sums-of-Squares (SOS) Programming

Proving that $\dot{V} < 0$ on the boundary $V = 1$ for a general nonlinear function $\boldsymbol{f}$ is famously difficult. We have an infinite number of states on the boundary to check. We circumvent this by restricting our dynamics $\boldsymbol{f}$ algorithms to polynomials (via Taylor expansion or substitution) and leveraging **Sums-of-Squares (SOS) Programming** [3, 16].

A polynomial $p(x)$ is a Sum of Squares if it can be written as $p(x) = \sum_i q_i(x)^2$ for some polynomials $q_i(x)$. If a polynomial is SOS, it is globally non-negative. Testing if a polynomial is SOS turns out to be equivalent to solving a Semidefinite Program (SDP)—a convex optimization problem that can be solved very efficiently.

We can reframe Theorem 7.1 using the S-procedure. To guarantee that $\dot{V}(\boldsymbol{x}, t) < 0$ whenever $V(\boldsymbol{x}, t) = 1$, we require:

$$-\dot{V}(\boldsymbol{x}, t) - L(\boldsymbol{x}, t)(1 - V(\boldsymbol{x}, t)) \quad \text{is SOS} \tag{7.3}$$

where $L(\boldsymbol{x}, t)$ is a strictly positive polynomial multiplier. If $V(\boldsymbol{x}, t) = 1$, the second term vanishes, and the SOS condition forces $-\dot{V} > 0$, achieving the invariance proof.

7.3.1 Implementation Sketch: Funnel Synthesis

The workflow below is the practical bridge from a nominal trajectory to a certifiable tube of allowable perturbations.

Require: Nominal trajectory $\boldsymbol{\gamma}^*(t)$, feedback law $\boldsymbol{u} = \boldsymbol{k}(\boldsymbol{x}, t)$, polynomial closed-loop model, time grid $\{t_k\}$.

Ensure: Time-varying Lyapunov matrices $\boldsymbol{P}(t_k)$ defining a verified funnel.

1: Linearize or polynomially approximate the closed-loop dynamics around $\boldsymbol{\gamma}^*(t)$.

2: Initialize $\boldsymbol{P}(t_k)$ from the transverse Riccati solution or another stabilizing quadratic form.

3: **for** each node t_k **do**

4: Define the candidate boundary $V_k(\boldsymbol{x}) = (\boldsymbol{x} - \boldsymbol{\gamma}^*(t_k))^\top \boldsymbol{P}(t_k)(\boldsymbol{x} - \boldsymbol{\gamma}^*(t_k))$.

5: Introduce SOS multipliers enforcing $-\dot{V}_k - L_k(1 - V_k)$ to be nonnegative on the boundary.

6: Add positive-definiteness constraints on $\boldsymbol{P}(t_k)$.

7: **end for**

8: **while** the SDP remains feasible **do**

9: Maximize the ellipsoid volume or a surrogate such as $-\log\det \boldsymbol{P}(t_k)$ across the grid.

10: If infeasibility appears, shrink the candidate funnel, refine the polynomial model, or shorten the horizon.

11: **end while**

12: Interpolate the verified $\boldsymbol{P}(t_k)$ between nodes to obtain a continuous funnel approximation.

13: **return** The verified funnel family $\mathcal{F}(t) = \{\boldsymbol{x} \mid V(\boldsymbol{x}, t) \leq 1\}$

Algorithm 7.1: Trajectory-Centered Funnel Synthesis

7.4 Computing the Funnel for the Golf Swing

Let us return to the double-pendulum golf model. The reference trajectory $\boldsymbol{\gamma}^*(t)$ from Chapter 6 gives nominal kinematics. The transverse LQR from Chapter 4 gives a feedback matrix $\boldsymbol{K}(t)$ designed to stabilize selected deviations. The goal is to compute a conservative funnel around this sequence and state the model assumptions under which the closed-loop system remains inside it.

We define our candidate Lyapunov function as a time-varying quadratic form:

$$V(\boldsymbol{x}, t) = (\boldsymbol{x} - \boldsymbol{\gamma}^*(t))^\top \boldsymbol{P}(t)(\boldsymbol{x} - \boldsymbol{\gamma}^*(t)) \tag{7.4}$$

where $\boldsymbol{P}(t)$ is a positive definite matrix that changes continuously along the trajectory. The funnel boundary is the time-varying ellipsoid where $V = 1$. By altering $\boldsymbol{P}(t)$, we alter the shape and orientation of the funnel.

Remark 7.2. Value Functions and Hamilton-Jacobi-Bellman Theory

The Lyapunov function $V(\boldsymbol{x}, t)$ in funnel synthesis is deeply connected to the value function from dynamic programming, and more broadly to the **Hamilton-Jacobi-Bellman (HJB) equation** in optimal control. The value function $J^*(\boldsymbol{x}, t)$ represents the optimal cost to reach the terminal constraint from state $\boldsymbol{x}$ at time t. In the deterministic setting of Agrachev and Sachkov's framework [1], the value function satisfies the HJB partial differential equation:

$$\frac{\partial J^*}{\partial t} + \min_{\boldsymbol{u}} \left[L(\boldsymbol{x}, \boldsymbol{u}) + \nabla_{\boldsymbol{x}} J^* \cdot \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \right] = 0. \tag{7.5}$$

The level sets of $J^*(\boldsymbol{x}, t) = c$ define the attainable set from which cost c can still be incurred to reach the target. The funnel synthesized via Lyapunov methods is a conservative approximation of this optimal value function: it guarantees invariance via the $\dot{V} < 0$ condition, making it a certificate of feasibility. Computing the exact value function would require solving the HJB equation, which is generally difficult; the Lyapunov/SOS approach trades exactness for verifiability.

Example 7.1. Volume Maximization via SDP

We transcribe the continuous time t into discrete nodes t_k, just as we did in direct collocation. At each node, we set up an SDP that seeks to *maximize* the volume of the ellipsoid defined by $\boldsymbol{P}(t_k)$, subject to the SOS invariance constraints connecting it to adjacent nodes.

The optimizer actively computes the exact structural limits of the golf swing's passive dynamics. It discovers that early in the downswing, the funnel can be incredibly "wide" (forgiving), but as centrifugal forces build, the SOS constraints force $\boldsymbol{P}(t_k)$ to become highly elongated along the unactuated dimensions in order to maintain the invariance guarantee. The resulting geometry is a true numerical portrait of athletic consistency.

7.5 Summary

Concept	**Role in Funnel Synthesis**
Lyapunov Function	Time-varying scalar $V(\boldsymbol{x}, t)$ that measures distance from the reference trajectory; boundary $V = 1$ defines the funnel surface.
Positive Invariance	Mathematical property ensuring that $\boldsymbol{x}(0) \in \mathcal{F}(0)$ implies $\boldsymbol{x}(t) \in \mathcal{F}(t)$ for all $t \in [0, t_f]$.
Sums-of-Squares (SOS)	Convex optimization technique to verify that $\dot{V} < 0$ on the funnel boundary, eliminating the need for numerical sampling.
Semidefinite Program	Optimization problem solved to compute the time-varying matrix $\boldsymbol{P}(t)$ that maximizes funnel volume subject to SOS constraints.
Transverse LQR	Provides initial feedback gains $\boldsymbol{K}(t)$ for stabilizing deviations orthogonal to the reference trajectory.
Funnel Synthesis Workflow	Practical sequence of polynomial approximation, SOS certification, and ellipsoid-volume maximization around the nominal trajectory.

In Chapter 8, we will explore how to make these funnels robust to temporal variation by migrating out of the time domain entirely and into the phase domain.

Chapter 8

Phase-Variable Control

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

Time is a river that carries us forward. Phase is the boat we build to navigate it.

In Plain Language 8.1. Ditching the Clock

If a robot is programmed to walk across a room in exactly 10 seconds, and you push it backwards halfway through, a naive time-tracking controller can interpret the lag as a large positional error. A human, by contrast, often organizes walking around physical milestones: "When my left foot hits the ground, swing my right leg forward." This is called using a "phase variable" instead of clock time. A phase variable is a physical marker (like the angle of the left leg) that tells the system where it is in the motion. The benefit is not automatic robustness; it depends on choosing a marker that remains observable, monotone, and well behaved over the operating region.

8.1 Spatial Paths vs. Temporal Execution

One common limitation of direct time-tracking architectures, when applied to biomechanics or advanced robotics, is their rigid adherence to the clock. If you construct a time-varying trajectory $\boldsymbol{\gamma}^*(t)$ for a robotic arm and command a tracking controller to follow it, the controller evaluates the error as $\boldsymbol{e}(t) = \boldsymbol{x}(t) - \boldsymbol{\gamma}^*(t)$.

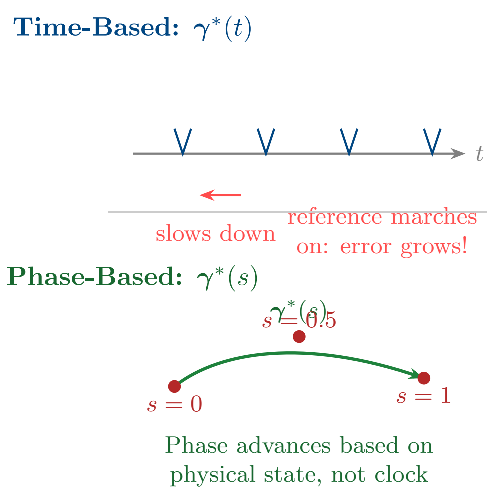
If the robot hits an unmodeled friction patch and slows down, the reference trajectory $\boldsymbol{\gamma}^*(t)$ continues advancing. The controller can calculate a large position error not because the robot has left the intended path, but because it is behind schedule. Depending on gains, actuator limits, and saturation handling, this can produce excessive catch-up commands.

Key Idea 8.1. The Autonomy of Time

A system executing a motion can often be controlled more safely when progress is measured by state-dependent phase rather than by clock time alone. If the system is impeded, the reference can slow down. The mathematical task is to decouple the spatial geometry of the motion from its temporal execution by replacing chronological time t with a locally monotone **phase variable** s.

Instead of parameterizing our optimal trajectory as $\boldsymbol{\gamma}^*(t)$, we parameterize it as a function of the phase: $\boldsymbol{\gamma}^*(s)$. The phase $s \in [0, 1]$ represents the percentage of motion completed. For a walking robot, s could be the forward position of the hip. For a golf swing, s could be the angular position of the shoulder relative to the ball. The reference trajectory is now a spatial curve with a separately modeled timing law.

This construction is publication-safe only after the phase variable's domain, monotonicity condition, and reset behavior are specified. This is the same practical issue that appears in maneuver regulation, transverse linearization, and hybrid zero dynamics [12, 24, 19].



Definition 8.1. Phase Variable

A phase variable $s(\boldsymbol{x})$ is a continuously differentiable scalar function of the state $\boldsymbol{x}$ such that its time derivative is strictly positive along all feasible trajectories inside the nominal funnel:

$$\dot{s}(\boldsymbol{x}) > 0 \quad \text{for all } \boldsymbol{x} \in \mathcal{F}. \tag{8.1}$$

8.2 Virtual Constraints

If the goal is no longer to match a time sequence but rather to conform to a spatial path defined by $s(\boldsymbol{x})$, we can express the control objective as a set of holonomic constraints on the state. However, because these constraints are not mechanical linkages but rather enforced by software, they are termed **Virtual Constraints** [24]. The theory of virtual constraints and hybrid zero dynamics was developed primarily in the context of bipedal locomotion by Westervelt, Grizzle, and collaborators.

Let our control objective be to drive the state $\boldsymbol{x}$ to the invariant manifold $\mathcal{Z}$ defined by:

$$\mathcal{Z} = \{\boldsymbol{x} \mid \boldsymbol{x} - \boldsymbol{\gamma}^*(s(\boldsymbol{x})) = \boldsymbol{0}\}. \tag{8.2}$$

In Plain Language 8.2. Mathematical Reminder: The Invariant Manifold

Recall our manifold definition from Volume 0: a curved geometric surface. The **Virtual Constraints** physically force the system to stick to a very specific, lower-dimensional "submanifold" embedded within the full State Space. We call this an **Invariant Manifold** because once the feedback controller forces the system onto this surface, the system's own natural physics (combined with the controller) ensures it never leaves it. The system "slides" perfectly along this curved surface toward the goal.

By using partial feedback linearization (Chapter 5), we can design a synthetic input $\boldsymbol{v}_a$ that drives the actuated degrees of freedom $\boldsymbol{q}_a$ onto their nominal surfaces proportional to the phase:

$$y = \boldsymbol{q}_a - \boldsymbol{\gamma}_a^*(s(\boldsymbol{x})) \to 0. \tag{8.3}$$

As $y \to 0$, the system converges toward the virtual constraints. Crucially, the rate at which it progresses along the path is entirely governed by $\dot{s}$, which is itself determined by the current momentum and state of the system, not a stopwatch.

8.3 Hybrid Zero Dynamics

When the virtual constraints are perfectly satisfied ($y = 0$), the entire complexity of the n-dimensional state space collapses down into a highly reduced dynamical system that evolves strictly along the manifold $\mathcal{Z}$. This is the **Zero Dynamics** of the system.

For systems that undergo discrete impacts (a foot hitting the ground, or a golf club striking a ball), the system evolves via continuous differential equations until an impact surface is triggered, followed by an instantaneous jump map (a reset) dictated by conservation of momentum.

Principle 8.1. Hybrid Zero Dynamics (HZD)

A control architecture achieves Hybrid Zero Dynamics if the invariant manifold $\mathcal{Z}$ (the zero dynamics surface) is invariant not just under the continuous flow of the differential equations, but also invariant under the discrete impact map. That is, an impact starting on the manifold $\mathcal{Z}$ must immediately map to a new state that is also on $\mathcal{Z}$.

By enforcing HZD, we can design rhythmic walking gaits that are astonishingly robust. If the robot trips, its phase slows down, but the virtual constraints remain active, keeping the limbs on the correct spatial path until the next footfall resets the sequence. The temporal sequence arises naturally from the interplay of gravity and momentum.

8.4 Application: The Golf Downswing

While the golf swing is not a periodic walking gait, a phase-variable approach can still be useful if the selected marker remains monotone through the interval being modeled. A candidate phase might be shoulder angle, hand path progress, or another measured progress coordinate; none is automatically valid outside the stated model. The target event is impact, not a universal manifold $s = 1.0$ independent of the coordinate choice.

By structuring a model through virtual constraints, the analysis can separate spatial deviations from timing delays. A transverse feedback law (Chapter 4) can correct selected spatial errors, while the phase variable schedules arm actuation (u_1) relative to measured progress rather than clock time. This can improve timing robustness, but it does not guarantee perfect passive impact dynamics; impact quality still depends on state estimation, actuator limits, contact modeling, and the validity of the chosen phase coordinate.

8.5 Summary

Concept	**Definition & Role**
Phase Variable	Scalar function $s(\boldsymbol{x})$ that increases monotonically along trajectories and decouples spatial path geometry from temporal execution.
Virtual Constraints	Control objective of the form $\boldsymbol{x} - \boldsymbol{\gamma}^*(s(\boldsymbol{x})) = 0$ that drives the system onto a lower-dimensional invariant manifold.
Invariant Manifold	The manifold $\mathcal{Z} = \{\boldsymbol{x} \mid \boldsymbol{x} = \boldsymbol{\gamma}^*(s(\boldsymbol{x}))\}$ on which the system evolves when virtual constraints are satisfied.
Hybrid Zero Dynamics	The reduced dynamics of the system when constrained to $\mathcal{Z}$, including discrete impact maps that preserve manifold membership.
Partial Feedback Lin.	Control design that cancels nonlinearities on actuated DOFs while allowing unactuated DOFs to evolve freely on the manifold.

Chapter 9

Stochastic Trajectories & Motor Variability

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

Perfection is inherently fragile. Biological optimization seeks the strongest balance between intent and reality.

In Plain Language 9.1. The Harder You Try, The More You Miss

If you program a robot motor to apply 50 units of force, it applies exactly 50 units of force, every single time. Human muscles don't work like that. The harder you try to contract a muscle, the more "noisy" and jittery the actual force becomes. This is a massive problem for trying to swing a golf club as fast as humanly possible: the closer you get to your maximum physical effort, the more unpredictable your motion becomes. This chapter explains the math behind why smooth, controlled movements are more accurate than jerky, maximum-effort ones. We'll see that a truly "optimal" golf swing doesn't use 100% of your strength, because dialing it back slightly eliminates a huge amount of physical chaos, making the swing actually repeatable.

9.1 Signal-Dependent Noise

When a roboticist specifies a torque of 50 Nm, the motor delivers 50 Nm with a tiny, constant variance. Biological actuators do not work this way. Human muscles generate force by recruiting discrete motor units [13]. As higher forces are required, larger, more fast-twitch oriented motor units are recruited. This physiological reality manifests as a fundamental control challenge: **motor noise is proportional to the command signal** [11].

Definition 9.1. Signal-Dependent Noise (SDN)

A control input $\boldsymbol{u}(t)$ applied by a biological system is actually realized as a stochastic variable $\tilde{\boldsymbol{u}}(t)$:

$$\tilde{\boldsymbol{u}}(t) = \boldsymbol{u}(t) + \boldsymbol{W}\boldsymbol{u}(t) \cdot \boldsymbol{\epsilon}(t) \tag{9.1}$$

where $\boldsymbol{W}$ is a scaling matrix representing the coefficient of variation, and $\boldsymbol{\epsilon}(t) \sim \mathcal{N}(0, I)$ is white noise. That is, the variance of the applied force grows squarely with the magnitude of the ordered force: $\mathrm{Var}(\tilde{u}) = cu^2$.

This simple reality crumbles deterministic optimal control. If you compute an optimal trajectory that requires maximum muscular exertion, the trajectory will mathematically possess minimum variance only if noise is constant. Under SDN, maximum exertion guarantees maximum chaotic noise.

9.2 Minimum Variance Theory of Human Movement

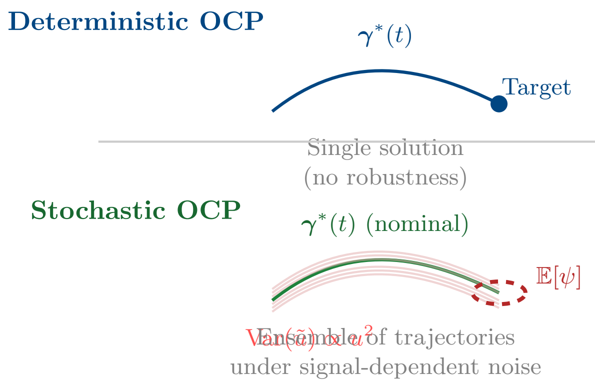
Why do humans move in smooth, bell-shaped velocity profiles rather than jerky "bang-bang" optimal control solutions when reaching for coffee?

One influential explanation is the **minimum jerk** theory of Flash and Hogan [8], which posits that the nervous system minimizes $\int \dddot{x}^2 \,\mathrm{d}t$ (the integral of squared jerk). This kinematic criterion successfully predicts many features of reaching movements. An alternative, physically grounded explanation comes from the **Minimum Variance Theory** of Harris and Wolpert [11], which derives smooth trajectories from the assumption of signal-dependent noise. Both theories remain active areas of research, and they are not mutually exclusive—minimum-jerk trajectories can emerge as a consequence of variance minimization under certain noise models.

Key Idea 9.1. Optimization for Repeatability

The human nervous system seeks to move in a way that minimizes the final endpoint variance evaluated across multiple trials. Under signal-dependent noise, this objective naturally selects for smooth trajectories with bounded control effort.

Under SDN, the only way to minimize endpoint variance (ensure you actually grasp the coffee cup smoothly) is to minimize sudden spikes in required actuator forces, because large forces inject huge amounts of uncertainty into the dynamics. Smooth trajectories inherently possess lower peak torque commands than rapid accelerations and decelerations. Smoothness is not the explicitly programmed objective; it is an emergent property of attempting to be consistently accurate under the unavoidable reality of signal-dependent noise.



9.3 The Speed-Accuracy Tradeoff in the Golf Swing

Fitts's Law [7] states that the faster you try to move to a target, the less accurate you become. In golf, we see the supreme manifestation of this phenomenon. The deterministic optimal control solution (Chapter 6) tells the solver to maximize input torque to achieve theoretically maximal clubhead speed.

However, if we introduce the Stochastic OCP (via Covariance Steering) from Chapter 6 and apply the realistic premise of SDN, the optimizer faces a stark tradeoff.

Applying maximum u_1 torque during the downswing injects enormous variance into the kinematic state. The unactuated club (θ_2) begins whipping through its zero dynamics, but its exact phase progression is now highly uncertain due to the noise flooding the arms.

To ensure the clubface squares up at the moving goal, the stochastic optimizer will *deliberately back off* the peak requested torque. It trades a small reduction in peak terminal performance for a substantial reduction in angular variance at the target manifold constraint. The optimal biological swing is mathematically proven to be a sub-maximal physical exertion when signal-dependent noise is accounted for.

9.4 Summary

Concept	Implication for Control
Signal-Dependent Noise (SDN)	Muscle noise variance proportional to command: $\mathrm{Var}(\tilde{u}) = cu^2$. Requires bounded control effort for repeatability.
Minimum Variance Theory	Explains smooth, controlled velocity profiles as solutions to endpoint error variance minimization under SDN.
Stochastic OCP	Optimal control formulation that minimizes $\mathbb{E}[\phi(\boldsymbol{x}(t_f))] + \lambda \cdot \mathrm{Var}[\psi(\boldsymbol{x}(t_f))]$.
Covariance Steering	Technique to propagate the covariance matrix $\boldsymbol{\Sigma}(t)$ alongside the nominal trajectory during optimization.
Speed-Accuracy Tradeoff	Formal manifestation of Fitts's Law: higher peak forces inject noise proportional to force squared.

The extension of the deterministic Pontryagin framework to stochastic systems is developed rigorously in Agrachev and Sachkov [1]. When the system is subject to signal-dependent noise (SDN), the traditional Hamiltonian maximization is supplemented by terms encoding risk: the optimizer must maximize expected performance while managing the variance of terminal conditions under noise propagation. This leads to **risk-sensitive control** problems where the cost includes an entropy-like term penalizing uncertainty.

For the golf swing under SDN, the stochastic extension of the attainable set framework tells us that not all nominally reachable configurations are equally *robustly* reachable. A configuration in the interior of the deterministic attainable set may lie on the boundary of the stochastic attainable set if noise-driven perturbations commonly push trajectories away from it. By contrast, configurations that are "stable" with respect to noise (in the direction of the control Lie algebra) remain deeply inside the stochastic attainable set. The funnel computed in Chapter 7 with stochastic cost weighting thus encodes not just reachability, but **robust attainment**—guarantee of success under unavoidable noise.

9.5 Summary

Concept	**Implication for Control**
Signal-Dependent Noise (SDN)	Muscle noise variance proportional to command: $\mathrm{Var}(\tilde{u}) = cu^2$. Requires bounded control effort for repeatability.
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Chapter 10

Learning to Move

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

The first swing establishes the manifold. The ten-thousandth swing perfects the funnel.

In Plain Language 10.1. Practice Makes Funnels

When you swing a golf club for the first time, your brain is guessing how much force to use, guessing the weight of the club, and guessing how your muscles will respond. The mathematical models we've built so far are just like that first swing—a really smart guess, but still a guess. Real physics (wind, friction, fatigue) is too messy to model perfectly. This chapter is about how systems *learn*. Instead of trying to calculate the perfect math equations from scratch every time, the system tries a movement, measures how badly it missed the target, and updates its "muscle memory" for the next try. Over thousands of repetitions, the brain (or the computer) learns the true physics of the motion without ever needing the exact equations. It builds a whole library of these learned motions, letting it instantly adjust to new situations without thinking.

10.1 Iterative Learning Control (ILC)

Trajectory optimization (Chapter 6) computes a reference motion strictly from a mathematical model. However, dynamic models of humans—and even robots—are invariably flawed. Frictional forces are unmodeled, inertias are estimated, and actuators possess hidden dynamics. If you execute a computed reference $\boldsymbol{\gamma}^*(t)$ open-loop, the physical system will likely fail to strike the moving target due to these discrepancies.

Iterative Learning Control (ILC) resolves this by leveraging the repetitive nature of tasks like a golf swing. ILC exploits the premise that while models are inaccurate, *the unmodeled dynamics are highly deterministic across repeated trials.*

Remark 10.1. ILC and the Orbit of Feasible Trajectories

From the perspective of geometric control theory as presented by Agrachev and Sachkov [1], each iterative trial of ILC represents an exploration within the orbit of feasible trajectories generated by the control Lie algebra. The true plant's dynamics define an actual (unknown) orbit in state space; the nominal model defines an approximate orbit. ILC learns the discrepancy between these orbits by iteratively modulating the feedforward command. The convergence of ILC is thus a convergence toward the true orbit boundary—the set of trajectories reachable on the actual (unknown) plant. Persistent excitation conditions required for identifiability of model parameters correspond to activating the control Lie algebra sufficiently to explore all dimensions of the reachable subspace. In other words, ILC's success in learning motion requires that the control inputs, applied across trials, persistently excite the full Lie algebra of admissible directions.

If a robot under-swings by 5 cm on the first attempt, the controller records this terminal error and updates the feedforward command $\boldsymbol{u}_{\text{ff}}$ for the *second* attempt. Over repeated trials, the system learns the inverse dynamics of the true physical plant without ever explicitly identifying the model parameters.

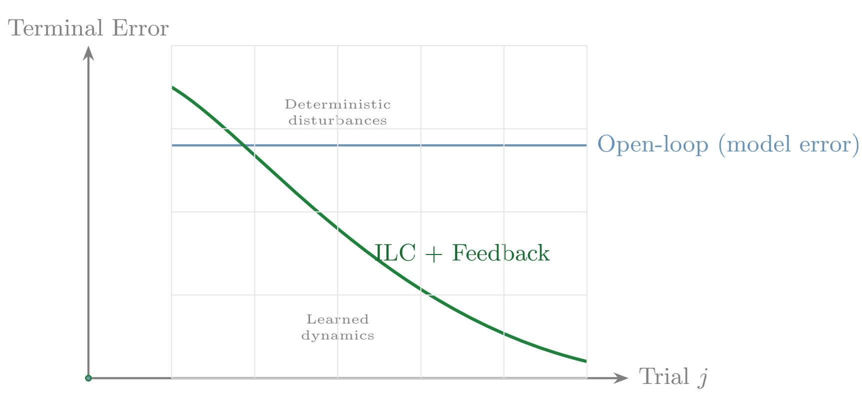
Definition 10.1. ILC Update Law

Let j denote the trial index. The feedforward command for trial $j+1$ is updated based on the command and the *transverse error* $\boldsymbol{\xi}$ from trial j:

$$\boldsymbol{u}_j(t) = \boldsymbol{u}_{\text{ff},j}(t) + \boldsymbol{K}_\perp(t)\boldsymbol{\xi}_j(t) \tag{10.1}$$

$$\boldsymbol{u}_{\text{ff},j+1}(t) = \boldsymbol{u}_{\text{ff},j}(t) + \boldsymbol{L}(t)\boldsymbol{\xi}_j(t) \tag{10.2}$$

where $\boldsymbol{L}(t)$ is a learning filter. This updates the nominal trajectory to iteratively cancel out deterministic disturbances.



Require: Nominal feedforward command $\boldsymbol{u}_{\mathrm{ff},0}(t)$, stabilizing feedback $\boldsymbol{K}_{\perp}(t)$, learning filter $\boldsymbol{L}(t)$.

Ensure: A learned command sequence with reduced trial-to-trial error.

```
for trial j = 0, 1, 2, ... do
Execute the closed-loop swing:
    u_j(t) = u_ff,j(t) + K_⊥(t) ξ_j(t).
Measure the transverse tracking error ξ_j(t) and the terminal miss at impact.
Filter the measured error to remove non-repeatable sensor noise.
Update the next feedforward command:
    u_ff,j+1(t) = u_ff,j(t) + L(t) ξ_j(t).
if the terminal error and update norm are below tolerance then
Stop learning.
end if
end for
return The learned feedforward command u*_ff(t)
```

Algorithm 10.1: Iterative Learning Control Around a Nominal Swing

10.2 Policy Search and Reinforcement Learning

While ILC updates the feedforward commands to zero-out deterministic tracking errors, **Policy Search** algorithms optimize the *shape of the funnel* itself.

Instead of computing the transverse LQR gains $\boldsymbol{K}_\perp(t)$ from an analytical Riccati equation (Chapter 4), we parameterize the controller as a policy $\pi_{\boldsymbol{\theta}}(\boldsymbol{x})$ and search the parameter space $\boldsymbol{\theta}$ to minimize the variance at the moving goal under realistic noise.

Reinforcement Learning (RL), specifically policy gradient methods, allows the agent to sample thousands of golf swings in simulation, continuously updating its policy parameters to maximize the expected reward (clubhead speed and squareness at impact). A heavily trained RL agent organically converges on the very principles we mathematically derived in earlier chapters:

1. It abandons rigid time-tracking and autonomously discovers phase-variable mapping.
2. It exploits the underactuated double-pendulum whipping motion to gain speed.
3. It artificially widens its "funnel" during the transition phase, realizing that overly tight feedback control here merely amplifies motor noise and spoils the impact constraints.

10.2.1 Evolutionary Policy Search

Not every learning architecture exposes gradients cleanly. If the policy contains contact logic, lookup tables, gait phase switches, or parameters chosen from expensive black-box simulation, **genetic** and related evolutionary methods become attractive [9, 22, 10]. Their role here is not to replace the geometry; it is to help search over controller families when differentiability is weak or deceptive.

Remark 10.2. Evolutionary Search as a Library Builder

The most natural place for evolutionary methods in this volume is the outer loop over *strategy parameters*: swing tempo, release timing, phase-variable thresholds, or shape parameters of a feedback policy. Once a promising family is found, the resulting motion should still be analyzed with the trajectory-optimization and funnel tools from the previous chapters.

10.3 Trajectory Libraries

A singular optimal trajectory $\boldsymbol{\gamma}^*$ and its corresponding funnel $\mathcal{F}$ are highly specific to a single objective. What if the golfer wishes to fade the ball, hit a draw, or execute a punch shot out of the rough?

Instead of re-running a massive NLP optimization for every scenario, biological systems and advanced controllers maintain **Trajectory Libraries**. A library consists of several distinct, pre-computed optimal trajectories and their corresponding local funnels.

When faced with a new task (e.g., striking the ball on an uneven lie), the system searches its library for the closest matching funnel. By interpolating between established funnels using techniques like LQR-Trees [23] or simple spatial blending, the system can instantly generate a stable, feasible motion without any online optimization.

Require: Task descriptor $\boldsymbol{z}$, library entries $\{(\gamma_i^*, \mathcal{F}_i, \boldsymbol{z}_i)\}$, local adaptation routine.

Ensure: A feasible warm start for the current task.

Compute similarity scores between the requested task $\boldsymbol{z}$ and each stored descriptor $\boldsymbol{z}_i$.

Select the best matching entry or a small elite subset.

if one funnel already contains the current initial condition and task variation **then**

Reuse that trajectory and controller directly.

else

Blend nearby trajectories or policy parameters to form a warm start.

Run a short local optimization or ILC phase to adapt the warm start to the new task.

end if

Verify that the adapted trajectory still lies inside a certified funnel near impact.

return The adapted trajectory/controller pair.

Algorithm 10.2: Trajectory-Library Retrieval and Adaptation

10.4 Summary

Concept	**Mechanism of Learning**
Iterative Learning Control (ILC)	Feedforward update: $\boldsymbol{u}_{\mathrm{ff},j+1} = \boldsymbol{u}_{\mathrm{ff},j} + \boldsymbol{L}(t)\boldsymbol{\xi}_j$. Learns deterministic model mismatch.
Policy Search	Direct optimization of feedback parameters $\boldsymbol{\theta}$ via gradient methods on sampled task performance.
Evolutionary Search	Population-based tuning of policy or trajectory parameters when gradients are unreliable or unavailable.
Reinforcement Learning (RL)	Learns both reference trajectories and feedback policies through trial-and-error in simulation or on real systems.
Trajectory Library	Pre-computed collection of optimal trajectories and funnels for multiple task variants; enables instantaneous adaptation via interpolation.
Model Identification	Learning the true plant dynamics $\boldsymbol{f}_{\mathrm{true}}(\boldsymbol{x}, \boldsymbol{u})$ to replace or refine the nominal model.

Chapter 11

Case Study: The Complete Golf Swing

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

A useful model makes its assumptions visible before it asks for trust.

In Plain Language 11.1. Putting It All Together

Everything we have developed so far can be assembled into a golf-swing model, but the case study should be read as a reproducible modeling program rather than as a proof of a universal swing law. Chapters 1–4 define the trajectory, state-space geometry, and transverse-error language. Chapters 5–8 add underactuation, trajectory optimization, funnels, and phase coordinates. Chapters 9–10 add signal-dependent noise and learning. This chapter shows how those pieces could be combined, and it states the assumptions that would have to be checked before stronger claims about expert performance are justified.

11.1 Unifying the "Control Is Motion" Framework

Throughout this text, we have reframed control around trajectory geometry and local transverse correction. We conclude with a hypothetical 15-DOF musculoskeletal golf-swing model that illustrates how the framework could be assembled into a testable analysis.

The objective is to maximize clubhead speed at the spatial intersection with the golf ball while controlling face orientation and respecting biological torque limits. The terminal time may be treated as a decision variable, but the formulation must still state admissible timing, contact, and phase-coordinate assumptions. Robustness is included through an explicit signal-dependent-noise model.

11.1.1 Phase 1: Defining the Geometry and Optimization

The system is heavily underactuated in this example. The wrists are treated as passive pin joints, which is a simplifying assumption used to expose inertial coupling rather than a complete anatomical model.

We formulate a **Stochastic Moving Target Optimal Control Problem** (Chapter 6). Direct collocation optimizes the trajectory $\boldsymbol{\gamma}^*(s)$ over a spatial phase variable s representing shoulder angle, provided that this phase coordinate remains monotone over the modeled interval (Chapter 8). The integral cost penalizes signal variance through local covariance propagation (Chapter 9).

In the illustrative model, the collocation solver satisfies the $\boldsymbol{B}^\perp$ annihilator constraint (Chapter 5) and may produce a smooth reference motion. A torque reversal before impact can exploit passive wrist dynamics, but claims about face squaring require explicit contact geometry, grip modeling, and validation against measured swings.

11.1.2 Phase 2: Funnel Synthesis and Robustness

With $\boldsymbol{\gamma}^*(s)$ computed, we analyze the trajectory using **Transverse Linearization** (Chapter 4) [12, 19]. We ignore errors along the tangent of the path only within the chosen phase model and construct a time-varying LQR feedback matrix to regulate selected transverse deviations.

Finally, we compute candidate bounds for this stability region using **Sums-of-Squares Programming** (Chapter 7) [16]. The SDP supplies a boundary function $V(\boldsymbol{x}, t) = 1$ that certifies invariance under the polynomial model, controller, and disturbance assumptions used in the computation.

Principle 11.1. Time-Varying Funnel Geometry

The computed funnel $\mathcal{F}(t) = \{\boldsymbol{x} \mid V(\boldsymbol{x}, t) \leq 1\}$ describes the model's certified region of attraction. The Lyapunov function $V(\boldsymbol{x}, t) = (\boldsymbol{x} - \boldsymbol{\gamma}^*(t))^\top \mathbf{P}(t)(\boldsymbol{x} - \boldsymbol{\gamma}^*(t))$ (Chapter 7) contains a time-varying matrix $\mathbf{P}(t)$ determined by the Riccati equation of the transverse linearization (Chapter 4).

During the early phases of motion, the funnel may be wider if the modeled dynamics tolerate larger transverse deviations without violating the invariant boundary. As the swing approaches transition and downswing, the funnel may tighten because the model becomes more sensitive to timing, phase, and actuation errors.

Any claim about a final 0.15 seconds, near-zero funnel diameter, or passive absorption of muscular noise must be supported by the actual SDP solution and sensitivity analysis. The funnel is a certificate for the stated model, not a general proof of athletic consistency.

This case study illustrates the intended use of the framework: define a motion objective, state the model assumptions, compute a trajectory, and certify the local control region. It does not replace empirical validation, and it should be read as a reproducible modeling program rather than a proof of a universal control language.

Glossary of Important Concepts

Articulated Body Algorithm A recursive $O(N)$ algorithm for computing the forward dynamics of a kinematic chain, widely used in physics engines.

Configuration Space The space (often a manifold) of all possible positions of a system.

Contraction Analysis A framework for analyzing the stability of trajectories with respect to each other, rather than with respect to an equilibrium point.

Control Contraction Metric (CCM) A Riemannian metric that can be used to synthesize stabilizing feedback controllers for nonlinear systems along arbitrary feasible trajectories.

Degrees of Freedom (DOF) The minimum number of independent coordinates required to completely specify the configuration of a system.

Drift The complete autonomous evolution of a declared effective plant when the declared control is zero, including inertia, gravity, Coriolis/centrifugal terms, passive elasticity, and damping.

Drift-Control Ratio (DCR) The ratio comparing modeled drift against bounded control authority in a declared acceleration or task-projected space: $\mathrm{DCR}_{W,\mathcal{U}}(\mathbf{x}) = \frac{\|W\mathbf{a}_{\mathrm{drift}}(\mathbf{x})\|_2}{\sup_{\mathbf{u}\in\mathcal{U}(\mathbf{x})}\|WB_a(\mathbf{x})\mathbf{u}\|_2+\varepsilon}$.

Euler Angles A 3-parameter representation of orientation that is subject to gimbal lock.

Exponential Map The operation that relates a Lie Algebra (representing velocity) to its corresponding Lie Group (representing position/orientation).

Flow The solution map of a differential equation smoothly moving points along vector fields in state space.

Gimbal Lock A coordinate singularity where a 3D rotation parameterization loses a degree of freedom.

Jacobian The matrix of partial derivatives relating joint velocities to task-space velocities.

Kinematics The study of motion geometry without considering the forces that cause it.

Lagrangian Mechanics An energy-based formulation of mechanics utilizing kinetic and potential energy to strictly determine the equations of motion.

Lie Algebra The mathematical space of velocities associated with a Lie Group, corresponding to its tangent space at the identity.

Lie Group A smooth manifold that is also a mathematical group, typically describing continuous symmetries or rigid body transformations (e.g., $SO(3)$, $SE(3)$).

Lyapunov Stability The classical definition of stability describing whether a system returns to an equilibrium point when perturbed.

Manifold A topological space that locally resembles Euclidean space, crucial for rigorous formulation of non-trivial state spaces.

Screw Axis A geometric representation of spatial motion combining a rotation axis and a translation along that axis, foundational for modern rigid body mathematics.

State Space The space of all possible states of a dynamical system, usually comprising both configuration and momentum/velocity.

Tangent Bundle The manifold consisting of all tangent spaces glued together, typically representing the full state space of mechanical systems.

Tangent Space The vector space of all possible velocities at a specific point on a manifold.

Twist The spatial velocity of a rigid body, composed of angular and linear velocity components, residing in the Lie Algebra.

Wrench The spatial force acting on a rigid body, combining torque and linear force.

Zero-Torque Counterfactual (ZTCF) family Model interventions setting the declared applied generalized-control channel to zero while preserving the declared effective plant. Includes pointwise samples, stitched traces, forward trajectories, and branched trajectories.

Zero-Velocity Counterfactual (ZVCF) The instantaneous zero-velocity, zero-control acceleration $\mathbf{a}_{\mathrm{ZVCF}}(\mathbf{q}, \mathbf{z}) = -\mathbf{M}(\mathbf{q})^{-1}\mathbf{h}(\mathbf{q}, \mathbf{0}, \mathbf{z}_0; \mathbf{p})$ isolating configuration-dependent loads without velocity-dependent or control contributions.

Further Reading and References

This text is a primer and introductory guide designed to construct an intuitive and unified framework. The formal depths of modern geometric mechanics and nonlinear control are actively pioneered and formalized across several seminal texts. For the reader eager to pursue more mathematically rigorous treatments, we strongly recommend the following resources:

Geometric Control and Robotic Manipulation

- **A Mathematical Introduction to Robotic Manipulation** [18]. The definitive textbook bridging Lie group theory with classical robotics, establishing the Product of Exponentials (PoE) standard.
- **Geometric Control of Mechanical Systems** [5]. An authoritative resource on the differential geometry of mechanical control systems, including connections and geodesics.
- **Robot Modeling and Control** [21]. A highly accessible and rigorous text covering everything from basic Euler-Lagrange forms to robust robotic control.

Advanced Rigid Body Dynamics

- **Rigid Body Dynamics Algorithms** [6]. To truly implement physics engines, Featherstone's Spatial Vector notation and the derivation of $O(N)$ ABA are unmatched.

Nonlinear Systems and Contraction

- **Applied Nonlinear Control** [20]. Slotine's classic text providing the most intuitive treatment of sliding mode and robust nonlinear stability.
- **Contraction Theory for Dynamical Systems** [4]. The modern codification of contraction bounding and metrics for trajectory stability.

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